

Prior-Conditioned Gaussian Discriminants for Generalizable AI-generated Image Detection

Shashank Kotyan, Makoto Shing, Yuki Imajuku, Rujikorn Charakorn and Tarin Clanuwat

Sakana AI, Tokyo, Japan

Diffusion-based generators have made synthetic images ubiquitous, but detectors often fail under simultaneous shifts in generator, prompt/style, and source-domain. We study AI-generated image detection as a transfer system described by training prior, frozen encoder feature space, and decision rule, and ask when classifier head training adds value beyond what is already separable in modern features. As a controlled diagnostic, we fit a prior-conditioned Gaussian discriminant ladder: closed-form heads built from first- and second-order feature statistics under nested covariance assumptions. On Percept-Lens, a unified protocol over 39 public datasets (7.1 million images), the best rung is frequently competitive with, and sometimes exceeds, released AI-generated image detector heads when matched on both prior and encoder. We further quantify strong sensitivity to the training prior, data-efficiency of moment-based heads, and representation dependence of Gaussian shift metrics, motivating (prior, encoder, head)-level reporting and stronger analytical baselines for AIGI transfer.

Keywords: AI-generated image detection; distribution shift; Gaussian discriminant analysis; few-shot transfer; transfer learning.

Contents

1	Introduction	2
2	Related Works	3
3	Preliminaries	4
4	Prior-Conditioned Gaussian Discriminants as Few-Shot Discriminant Heads	4
5	Investigating Generalization with Percept-Lens suite	7
6	Conclusion	13
A	On the Computational Paradigm and Practical Utility of Gaussian Discriminants	21
B	Proofs of Theoretical Propositions	21
C	Analytical Forms of the Gaussian Discriminant Ladder	25
D	Details about existing datasets in our Percept-Lens evaluation suite	26
E	Extended Investigation of Generalization with Percept-Lens suite	27
F	Threshold-Free (AUC) Evaluation on Mixed Datasets in Percept-Lens suite	31
G	Class-Conditional Gaussianity Diagnostics of Evaluation Features	39

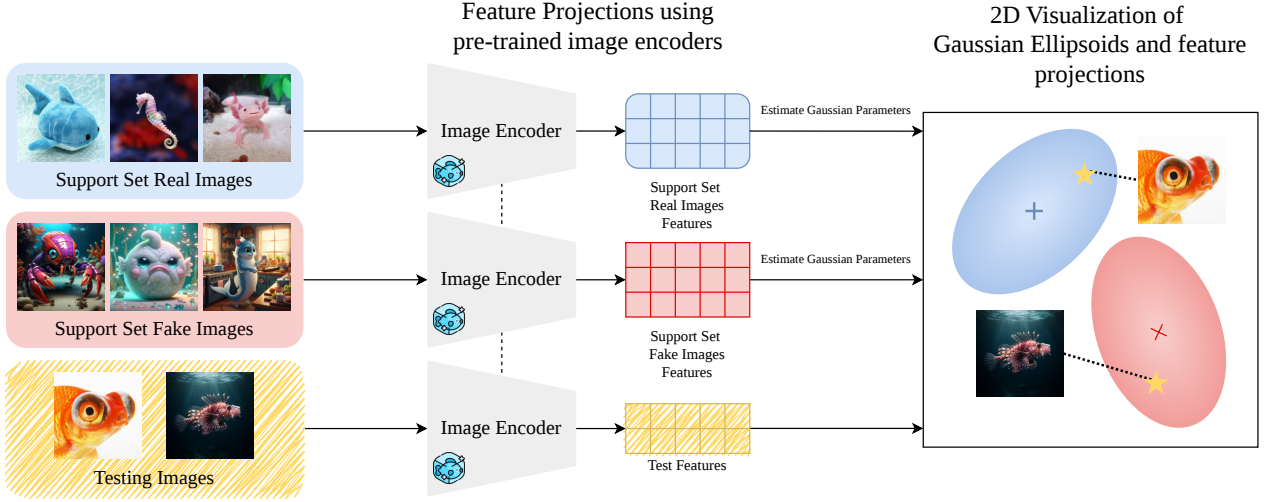


Figure 1 | **Training-prior-conditioned Gaussian discriminants in a frozen feature space.** A frozen image encoder ϕ maps images x to embeddings z . Given a (possibly few-shot) support set S drawn from a *training prior*, we estimate class-conditional moments (μ_c, Σ_c) and instantiate a ladder of *classical* closed-form heads under nested covariance assumptions. By keeping the encoder and prior fixed, the ladder isolates what the decision rule contributes.

1. Introduction

Diffusion-based generative models (MidJourney, 2022; Ramesh et al., 2021; Rombach et al., 2022; Saharia et al., 2022) have made high-quality AI-generated imagery widely accessible. These models have practical benefits, but they also complicate trust in visual media and enable misuse of synthetic content (Bushard, 2023; Daniel, 2024; McCarthy, 2023). As a result, AI-generated image (AIGI) detection has become a core recognition problem with direct implications for safety and privacy.

Modern detectors must transfer across shifts in *both* generator family and image-source domain. In practice these shifts co-occur (new generators, new prompting styles, new platforms, and post-processing), and performance can collapse even when in-distribution metrics look strong. Moreover, collecting large-scale labeled data from newly emerging generators is often expensive or infeasible. Together, these conditions make AIGI detection a concrete *transfer* / *low-shot recognition* problem, where a deployed detector must operate under joint shift and sometimes with only a small support set for calibration.

Existing evaluations (Baraldi et al., 2025; Chen et al., 2024; Zhu et al., 2023) typically probe only part of this space, frequently reusing narrow prompt regimes and closely related sources. Such evaluations can conflate genuine generalization with prior-specific shortcuts. We take a stricter, system-level view. A reported number is a property of a *system* consisting of a training prior, a frozen image encoder, and a decision rule on top. Here, *training prior* denotes the data distribution induced by a public training dataset, such as CommunityForensics (Park and Owens, 2025); it is not a Bayesian prior over model parameters. If generalization fails, it is not obvious whether the bottleneck is the representation, the head optimization, or the prior itself.

To isolate the decision-rule factor while keeping the prior and representation fixed, we fit *training-prior-conditioned Gaussian discriminants*, which are classical closed-form heads obtained from first- and second-order statistics of frozen encoder features (Figure 1). We organize these rules into a *Gaussian discriminant ladder* spanning nested covariance assumptions (isotropic, diagonal, shared, class-specific). Because each rung has an analytic solution, the ladder is a reproducible, hyperparameter-light baseline. The identity of the best-performing rung also indicates which low-order feature statistics transfer across generator and domain shifts. Although Gaussian discriminants are classical,

we argue that they are underused as diagnostic baselines in AIGI detection, and we find that they can be competitive under matched (prior, encoder) conditions.

Empirically, across 39 public test sets (7.1 million images), at least one rung is often competitive with trained heads under *matched* priors and frozen encoders. As a retrospective diagnostic upper bound, the best rung can even surpass released heads in several settings. Our decomposition is not fully symmetric across the three factors. The most direct intervention is a matched head replacement, while the encoder and prior experiments are controlled one-factor sweeps. We use the procedure as a source-side audit. If a trained head does not outperform the best rule in the ladder on the same final representation and support prior, classifier training is unlikely to be the source of OOD transfer gains. If it succeeds, it signals structure beyond second moments.

Contributions.

Prior-conditioned Gaussian discriminant ladder (diagnostic baseline). We instantiate a ladder of *classical* closed-form Gaussian heads on frozen encoder features, varying covariance assumptions (isotropic $\rightarrow$ diagonal $\rightarrow$ shared full $\rightarrow$ class-specific full). The best-transfer rung provides an interpretable diagnostic of which low-order feature statistics are preserved under shift.

Post-hoc controlled evaluation under matched (prior, encoder) conditions. On Percept-Lens, a unified evaluation *protocol* built entirely from existing public datasets, we compare released AI-generated image detector heads against the ladder using the same training prior and the same frozen encoder features. This isolates when trained heads add value beyond low-order geometry.

Empirical drivers of generalization under joint shift. We quantify sensitivity to the training prior under a fixed encoder, show that moment-based heads can adapt with few labeled samples on strong frozen encoders, and demonstrate that Gaussian shift metrics are strongly representation-conditioned.

2. Related Works

2.1. AI-generated Image (AIGI) detection under Distribution Shift

The literature on AIGI detection spans artifact-driven detectors (He et al., 2021; Wang et al., 2020; Yang et al., 2019) and representation-based detectors (Baraldi et al., 2025; Ojha et al., 2023; Park and Owens, 2025). Early work focused on GAN-era artifacts and domain-specific settings such as face forgeries (He et al., 2021; Wang et al., 2020; Yang et al., 2019). Diffusion models increased photorealism and prompt diversity, and exposed a persistent failure mode: detectors trained on narrow synthetic prior exploit shortcuts that do not transfer across generators or domains (Baraldi et al., 2025; Chen et al., 2024; Xiao et al., 2025; Zhu et al., 2023). Chen et al. (2024) and Park and Owens (2025) broaden training priors by scaling generator coverage. Ojha et al. (2023), Baraldi et al. (2025), and Zhou et al. (2025a) instead use foundation-model features and lightweight heads. Yang et al. (2026) adjust detector logits or thresholds after training under shift. Our intervention instead replaces the classifier in feature space while matching the prior and encoder. Thus our emphasis is not another trained detector, but what is already separable in frozen encoder space and what the trained head adds under the *same* prior. This is closely related to the observation that many OOD detectors can be expressed as generative scoring rules on features, but the implications for AIGI detection under joint shift have not been carefully isolated.

2.2. Analytical Gaussian Discriminants and Feature-Space Scoring

Gaussian discriminant analysis and Mahalanobis scoring are classical tools for uncertainty and OOD detection (Lee et al., 2018). Wu et al. (2025b) connect few-shot prototypical inference to Euclidean

nearest-centroid rules, while [Yan et al. \(2025\)](#) explore orthogonal subspace decompositions that preserve pretrained structure under adaptation.

We build on these ideas, but use them as a *diagnostic ladder*: isotropic, diagonal, tied, and class-specific covariance assumptions correspond to increasing geometric expressivity. The decision rules themselves are not new. Our contribution is to condition them on explicit training priors and use the ladder covariance model to interpret what transfers, and what does not, across generator and domain shifts.

3. Preliminaries

3.1. Problem setup

We study binary detection of AI-generated images. Given a training set $\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^n$, where x_i are images and $y_i \in \{0, 1\}$ are labels for real ($y = 0$) and synthetic ($y = 1$) images, the goal of AIGI detection is to learn a classifier that can accurately predict the label of a previously unseen image.

3.2. Feature Extraction and Trained Classifiers

We use a frozen image encoder $\phi : \mathcal{X} \rightarrow \mathbb{R}^D$ from a pre-trained foundation model to map each image x to a D -dimensional feature vector $z = \phi(x)$. A standard approach trains a discriminative head, such as a linear probe, on these features by minimizing a loss function (e.g., binary cross-entropy) via gradient descent: $\min_{w,b} \sum_{i=1}^n \text{CE}(\sigma(w^\top z_i + b), y_i)$ where $\sigma(\cdot)$ is the sigmoid activation function and $w \in \mathbb{R}^D$ and $b \in \mathbb{R}$ define a separating hyperplane.

More complex models, such as multi-layer perceptrons (MLPs), can learn non-linear decision boundaries. This process iteratively searches for an optimal boundary based on the training data. Although powerful, these methods can overfit to artifacts specific to the generative models in the training set, potentially limiting generalization to unseen generative processes ([Park and Owens, 2025](#); [Yan et al., 2025](#)).

We compare (i) *trained* discriminative detectors that optimize a supervised objective on a given prior and (ii) *Gaussian discriminant rule* classifiers fitted on frozen features (Section 4). The audit applies to released systems that expose the feature vector used by a classifier, including frequency/statistical detectors ([Frank et al., 2020](#); [Yan et al., 2024](#)), reconstruction-based detectors ([Cazenavette et al., 2024](#); [Chen et al., 2024](#)), LoRA-adapted detectors ([Yan et al., 2025](#)), or end-to-end detectors ([Baraldi et al., 2025](#); [Wang et al., 2020](#)) after their final representation is fixed. Methods that expose only a scalar anomaly score can still be evaluated by Percept-Lens as complete detectors, but their internal head cannot be replaced by the ladder. The split is diagnostic: if a trained head underperforms a closed-form rule under OOD shift, the trained head’s decision surface and/or training prior is implicated rather than only the encoder capacity.

4. Prior-Conditioned Gaussian Discriminants as Few-Shot Discriminant Heads

We use Gaussian class-conditional models as a *controlled diagnostic* of what information is already separable in a frozen feature space. Motivated by classical generative classification and recent analyses of contrastive representations using mixture models ([Bansal et al., 2025](#)), we approximate

$$p(z \mid y = c) \approx \mathcal{N}(\mu_c, \Sigma_c), \quad c \in \{0, 1\}. \quad (4.1)$$

Given a support set $\mathcal{S}$ drawn from a training prior, $\mathcal{S} \subseteq \mathcal{D}_{\text{train}}$, we estimate (μ_c, Σ_c) by sample moments. We also define the pooled covariance $\Sigma_p = \frac{(n_0-1)\Sigma_0 + (n_1-1)\Sigma_1}{n_0+n_1-2}$, used when assuming a shared

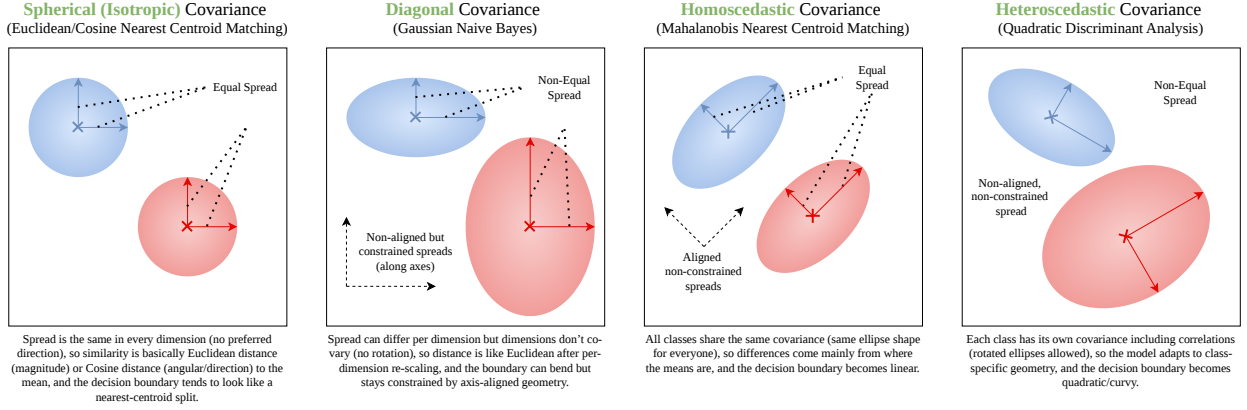


Figure 2 | **Gaussian discriminant ladder as nested covariance assumptions.** Each rung is a classical generative classifier obtained by restricting the class-conditional covariance: isotropic (Euc-/Cos-NCM), diagonal (GNB), shared full covariance (Mah-NCM), or class-specific full covariance (QDA). This nested family makes it possible to diagnose how much transferable signal is captured by first- and second-order feature statistics.

covariance structure. In high-dimensional embeddings, covariance inversion can be ill-conditioned. We therefore use standard regularization (diagonal loading and shrinkage) when computing Σ^{-1} and log-determinants (Appendix Section D). The Gaussian assumption is diagnostic rather than literal. Appendix Tables 23 and 24 shows mostly near-Gaussian marginal summaries on PE-Core features, but also clear failures such as FourierSpectrumDiscrepancies (Dzanic et al., 2020) and DiffusionForensics-fake (Wang et al., 2023). Multimodal or heavy-tailed regimes are exactly where trained non-linear heads may add value.

This yields a ladder of *classical* discriminant rules in closed form, and the best-performing covariance assumption becomes an interpretable indicator of which feature statistics transfer across domains, bypassing iterative training. (The same estimation extends to N -way K -shot classification. We focus on binary AIGI detection.)

In this article, we consider Cosine Nearest Centroid Matching (Cos-NCM), Gaussian Naive Bayes (GNB), Mahalanobis Nearest Centroid Matching (Mah-NCM), and Quadratic Discriminant Analysis (QDA) as Gaussian discriminants. We also include squared Euclidean nearest-centroid matching (Euc-NCM), which mirrors the inference rule used in prototypical-network detectors (e.g. Wu et al., 2025b) but is applied directly in the frozen encoder space without metric learning. Table 1 provides an overview of Gaussian discriminants in analytical form, while Figure 2 provides a visual explanation.

Euclidean Nearest Centroid Matching (Euc-NCM). Assuming an isotropic shared covariance $\Sigma_c = \sigma^2 I$, the Gaussian log-likelihood (up to additive constants) is proportional to the squared Euclidean distance to the class mean. Prediction reduces to nearest-centroid classification in the frozen feature space, matching prototypical inference without metric learning.

Cosine Nearest Centroid Matching (Cos-NCM). When features are ℓ_2 -normalized, isotropic Gaussian scoring is monotone in cosine similarity. Cos-NCM predicts the class whose centroid has the highest cosine similarity to the query.

Gaussian Naive Bayes (GNB). GNB assumes diagonal covariance $\Sigma_c = \text{diag}(\sigma_{c,1}^2, \dots, \sigma_{c,D}^2)$, which implies conditionally independent features. The assumption simplifies classification but ignores feature correlations.

Table 1 | **Gaussian discriminant ladder used as analytical baselines.** Each rung is a generative classifier obtained by restricting the class-conditional covariance (isotropic, diagonal, homoscedastic, or heteroscedastic), which determines the geometry of the induced decision boundary. We use this ladder to isolate head effects in a fixed frozen encoder feature space rather than to propose a new detector.

Gaussian Discriminant	Covariance Structure Assumption		Decision Boundary	Characteristic
Euclidean Nearest Centroid Matching (Euc-NCM)	Isotropic	$\Sigma_{c \in \{0,1\}} = \sigma^2 I$	Linear	Assumes spherical class distributions.
Cosine Nearest Centroid Matching (Cos-NCM)	Isotropic	$\Sigma_{c \in \{0,1\}} = \sigma^2 I$	Linear	Assumes spherical class distributions; decision is based on angular proximity, suitable for normalized embeddings.
Gaussian Naive Bayes (GNB)	Diagonal	$\Sigma_{c \in \{0,1\}} = \text{diag}(\sigma_{c,j}^2)$	Non-Linear	Simplest model; assumes feature independence, useful for identifying decorrelated representations.
Mahalanobis Nearest Centroid Matching (Mah-NCM)	Homoscedastic	$\Sigma_0 = \Sigma_1 = \Sigma_p$	Linear	Assumes classes share the same elliptical shape. It accounts for feature correlations by whitening the space, yielding a correlation-aware linear boundary.
Quadratic Discriminant Analysis (QDA)	Heteroscedastic	$\Sigma_0 \neq \Sigma_1$	Quadratic	Most general model; allows each class to have a unique elliptical shape and orientation, capturing complex separations.

Mahalanobis Nearest Centroid Matching (Mah-NCM). Mah-NCM assumes homoscedasticity, i.e., a shared covariance matrix for both classes $\Sigma_0 = \Sigma_1 = \Sigma_p$. The classification decision is based on the Mahalanobis distance $(z - \mu_c)^\top \Sigma_p^{-1} (z - \mu_c)$, which accounts for the correlations of the features by transforming the feature space into one in which the pooled covariance is the identity.

Quadratic Discriminant Analysis (QDA). QDA is the most general of the four classifiers, allowing each class to have its own full-rank covariance matrix ($\Sigma_0 \neq \Sigma_1$), thus assuming heteroskedasticity. The resulting quadratic boundary can adapt to classes with different shapes and orientations.

4.1. Connection to classical generative classification

The efficacy of these Gaussian discriminants in analytical form can be understood through the lens of Bayesian decision theory, which provides a formal basis for our geometric approach. Under the Gaussian assumption, they approximate or realize the Bayes-optimal classifier, which minimizes the probability of error. Full proofs are provided in the Appendix Section B.

Proposition 1 (Bayes-Optimal Classifier under Gaussian Assumption). *If the class-conditional densities are Gaussian, $p(z | y = c) \approx \mathcal{N}(\mu_c, \Sigma_c)$, and the class priors are equal, $P(y = 0) = P(y = 1) = \frac{1}{2}$, the Bayes-optimal decision rule is given by Quadratic Discriminant Analysis (QDA).*

Proposition 2 (Optimality under Homoscedasticity). *If the classes are additionally assumed to be homoscedastic (i.e., they share a common covariance matrix $\Sigma_0 = \Sigma_1 = \Sigma_p$), the Bayes-optimal decision rule simplifies to Mahalanobis Nearest Centroid Matching (Mah-NCM).*

Proposition 3 (Performance Stability under Bounded Drift). *Assume a homoscedastic classifier with shared covariance Σ and equal priors. If the statistical drift between the training and testing distributions is bounded by $\|\mu_c^{\text{test}} - \mu_c^{\text{train}}\|_2 \leq \epsilon_\mu$ and $\|\Sigma^{\text{test}} - \Sigma^{\text{train}}\|_F \leq \epsilon_\Sigma$, then $|\text{AUC}_{\text{test}} - \text{AUC}_{\text{train}}| \leq L(\epsilon_\mu + \epsilon_\Sigma)$ where L is the Lipschitz constant.*

Proposition 4 (Stability of the Fisher Margin under Distributional Drift). *Assume that both the training and testing domains are characterized by homoscedastic Gaussian parameters. If the distributional drift is bounded such that $\|\delta^{\text{test}} - \delta^{\text{train}}\|_2 \leq \epsilon$ and $\|\Sigma_p^{\text{test}} - \Sigma_p^{\text{train}}\|_F \leq \eta$, where $\delta = \mu_1 - \mu_0$, then the absolute change*

in the Fisher Margin $\mathcal{D}_\mu$ is bounded, to a first order, by $|\mathcal{D}_\mu^{\text{test}} - \mathcal{D}_\mu^{\text{train}}| \leq 2\|(\Sigma_p^{\text{train}})^{-1}\|_2\|\delta^{\text{train}}\|_2\epsilon + O(\eta)$.

These propositions should be read as *model-conditional*: they characterize the optimal decision rules if frozen features are well-approximated by Gaussian class-conditionals. They provide the theoretical foundation for our geometric analysis of generalization. Under this model, Quadratic Discriminant Analysis is Bayes-optimal in the heteroscedastic case (Proposition 1) and simplifies to a linear rule under a shared covariance (Mahalanobis-NCM; Proposition 2). Proposition 3 shows that, under the homoscedastic Gaussian model, the AUC of Mah-NCM is a monotone function of the Fisher margin $\mathcal{D}_\mu$. Proposition 4 bounds how $\mathcal{D}_\mu$ changes under drift in low-order statistics. Together, these results imply smooth degradation when class means and covariances drift. In this model, reliable generalization requires geometric alignment between training and testing data.

The linear decision boundary of Mah-NCM has the same functional form as a linear probe trained with cross-entropy. Both induce a single separating hyperplane in z -space. The key difference is optimization rather than representation. A neural network iteratively searches for the boundary, whereas Gaussian discriminants compute it analytically from the data’s first- and second-order statistics. Thus, when a trained linear head does not outperform Gaussian discriminants under matched priors and encoders, the result is consistent with the hypothesis that second-order feature geometry is already sufficient for separation. Consistent gaps instead suggest non-Gaussian or higher-order structure that the ladder cannot capture.

Limitations. By design, the Gaussian ladder is restricted to a Gaussian class-conditional approximation based on first- and second-order statistics. The result is a diagnostic baseline rather than a universal detector, one that leaves out higher-order, multimodal, or heavy-tailed cues from strong post-processing, unusual content, or generator-specific artifacts. In those regimes, trained non-linear heads or end-to-end adaptation can legitimately outperform the ladder.

5. Investigating Generalization with Percept-Lens suite

We use the Gaussian ladder to investigate which components of a detection system drive OOD performance under realistic distribution shift. We focus on the training prior, the frozen encoder, and the decision rule on top. To assess the reliability of current AI-generated image detectors under realistic generative variability, we begin by evaluating pre-trained models on the Percept-Lens test suite. These results help to localize whether failures are attributable to the training prior, the encoder feature space, or the trained decision surface, motivating more principled system-level investigations.

5.1. Evaluation Protocol and Data

Percept-Lens denotes a unified evaluation *protocol* constructed entirely from existing public datasets to stress-test AIGI detectors under joint distribution shift. Unlike settings where only the generator family changes while the underlying image source and prompt distribution remain fixed, Percept-Lens explicitly couples shifts in generator family, prompt/style, and acquisition pipeline (Figure 3). To avoid conflating in-distribution performance with transfer, the large training priors used for conditioning (e.g., CommunityForensics, GenImage, DRCT-2M, ELSA-D3) are treated as support-only and are not part of the evaluation suite.

Evaluation Datasets. The Percept-Lens suite aggregates 39 existing public datasets totaling 7.1 million images. It spans real-only, synthetic-only, and mixed regimes, with joint shifts in (i) generator family, (ii) prompt/style, and (iii) image-source domain and post-processing. Concretely, it combines in-the-wild real sources, synthetic-only prompt/style collections, and mixed forensic benchmarks spanning diverse generators and post-processing.

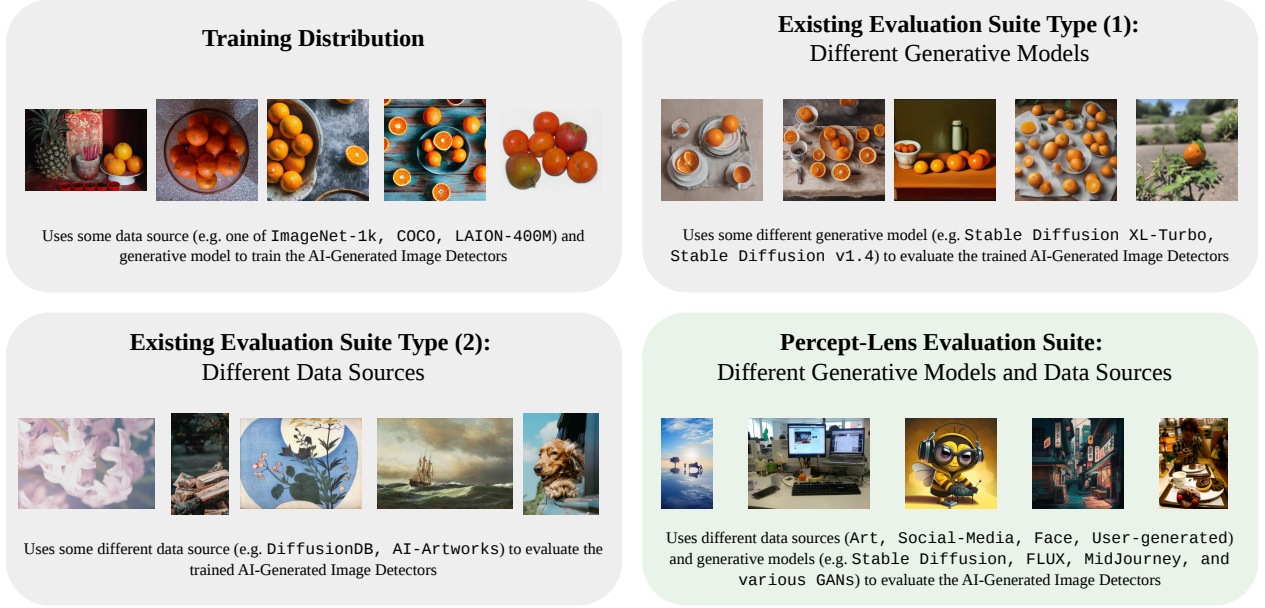


Figure 3 | **Percept-Lens stress-tests detectors under joint distribution shift.** Percept-Lens is an evaluation *protocol* built entirely from existing public datasets. Many AIGI evaluations vary either the generator family *or* the image source in isolation; we instead evaluate under *joint* shifts in generator family, prompt/style, and source domain, closer to deployment where these factors drift together.

Table 2 | **Closed-form Gaussian discriminants versus released AI-generated image detector heads under matched (prior, encoder) conditions.** Mean class accuracy CA (dataset-wise balanced accuracy where defined; Section 5.1), macro-averaged across Percept-Lens evaluation datasets. For each released detector checkpoint, we report (i) the released decision head (*Out-of-the-Shelf*) and (ii) closed-form baselines fitted on the *same frozen encoder features* using the corresponding public training prior: Euc-NCM (Wu et al., 2025b) and the best-performing rung of the Gaussian ladder. The full ladder is reported in Table 3. (Parentheses indicate the training subset reported by the original work when applicable.)

Detection Model	Out-of-the-Shelf	Euc-NCM (Wu et al., 2025b)	Gaussian ladder (best)
Trained with CNNSpot (Wang et al., 2020) (ProGAN Images based on LSUN)			
UnivFD (Ojha et al., 2023)	54.63%	50.98%	57.62% (Mah-NCM) (2.99% ↑)
AIDE (Yan et al., 2024)	56.41%	54.25%	62.76% (Mah-NCM) (6.35% ↑)
Trained with GenImage (Zhu et al., 2023) (Diffusion Model Images based on ImageNet-1k)			
AIDE (GenImage-SDv1) (Yan et al., 2024)	48.99%	54.53%	57.35% (Mah-NCM) (8.36% ↑)
Effort (GenImage-SDv1) (Yan et al., 2025)	72.58%	75.06%	76.82% (Mah-NCM) (4.24% ↑)
DRCT-UnivFD (Full GenImage) (Chen et al., 2024)	65.77%	66.08%	73.01% (Mah-NCM) (7.24% ↑)
AIDE (Full GenImage) (Yan et al., 2024)	54.98%	52.24%	64.64% (Mah-NCM) (9.66% ↑)
Trained with DRCT-2M (Chen et al., 2024) (Stable Diffusion Model Images based on COCO)			
DRCT-UnivFD (DRCT-SDv1) (Chen et al., 2024)	63.36%	65.36%	70.20% (QDA) (6.84% ↑)
DRCT-UnivFD (DRCT-SDv2) (Chen et al., 2024)	62.37%	66.31%	70.31% (QDA) (7.94% ↑)
Trained with ELSA-D3 (Baraldi et al., 2025) or CommunityForensics (Park and Owens, 2025) (Diffusion Model Images based on LAION-400M Dataset)			
CoDE-kNN (ELSA-D3) (Baraldi et al., 2025)	64.01%	63.63%	66.15% (Mah-NCM) (2.14% ↑)
CF-224 (CommunityForensics) (Park and Owens, 2025)	81.98%	78.53%	82.61% (Mah-NCM) (0.63% ↑)
CF-384 (CommunityForensics) (Park and Owens, 2025)	87.54%	82.65%	84.55% (Mah-NCM) (-2.99% ↓)
Frozen image encoder (no detection fine-tuning)			
PE-Core-bigG-14-448 (Bolya et al., 2025)	—	86.38%	94.46% (Mah-NCM)

Percept-Lens standardizes evaluation over existing public datasets to ensure head–prior–encoder comparisons are tested under a realistic mixture of shifts. The complete manifest of the dataset and the composition of the class are provided in Appendix Table 9.

Evaluation Protocol. For a training prior, detectors are evaluated on every dataset in the Percept-Lens test suite. We report a dataset-wise class accuracy that equals balanced accuracy when both classes are present and reduces to class-conditional accuracy on one-class sets. Concretely, for an evaluation dataset t with observed classes $C_t \subseteq \{0, 1\}$, we define Class Accuracy CA as

Table 3 | **Comparative Performance of Different Gaussian Discriminants.** Mean class accuracy CA (macro-averaged across Percept-Lens evaluation datasets) for each rung of the Gaussian discriminant ladder, conditioned on the same public training prior and fitted on the same frozen features as in Table 2. (Parentheses indicate the training subset reported by the original work; underlining denotes improvement over the corresponding released head as reported in Table 2.)

Detection Model	Cos-NCM	GNB	Mah-NCM	QDA
Trained with CNNSpot (Wang et al., 2020) (ProGAN Images based on LSUN Dataset)				
UnivFD (Ojha et al., 2023)	51.01%	48.44%	<u>57.62%</u>	48.23%
AIDE (Yan et al., 2024)	54.36%	<u>56.69%</u>	<u>62.76%</u>	<u>61.28%</u>
Trained with GenImage (Zhu et al., 2023) (Diffusion Model Images based on ImageNet-1k Dataset)				
AIDE (GenImage-SDv1) (Yan et al., 2024)	<u>54.23%</u>	<u>54.39%</u>	<u>57.35%</u>	<u>57.15%</u>
Effort (GenImage-SDv1) (Yan et al., 2025)	<u>74.44%</u>	<u>74.92%</u>	<u>76.82%</u>	<u>73.21%</u>
DRCT-UnivFD (Full GenImage) (Chen et al., 2024)	<u>67.97%</u>	<u>68.28%</u>	<u>73.01%</u>	<u>69.74%</u>
AIDE (Full GenImage) (Yan et al., 2024)	52.66%	53.92%	<u>64.64%</u>	<u>56.20%</u>
Trained with DRCT-2M (Chen et al., 2024) (Stable Diffusion Model Images based on COCO Dataset)				
DRCT-UnivFD (DRCT-SDv1) (Chen et al., 2024)	<u>65.16%</u>	<u>66.37%</u>	<u>65.21%</u>	<u>70.20%</u>
DRCT-UnivFD (DRCT-SDv2) (Chen et al., 2024)	<u>65.53%</u>	<u>66.69%</u>	<u>67.91%</u>	<u>70.31%</u>
Trained with ELSA-D3 (Baraldi et al., 2025) or CommunityForensics (Park and Owens, 2025) (Diffusion Model Images based on LAION-400M Dataset)				
CoDE-kNN (ELSA-D3) (Baraldi et al., 2025)	62.70%	65.98%	<u>66.15%</u>	64.00%
CF-224 (CommunityForensics) (Park and Owens, 2025)	78.98%	79.45%	<u>82.61%</u>	80.70%
CF-384 (CommunityForensics) (Park and Owens, 2025)	84.46%	80.01%	84.55%	79.52%
Frozen image encoder (no detection fine-tuning)				
PE-Core-bigG-14-448 (Bolya et al., 2025)	86.06%	87.26%	<u>94.46%</u>	86.15%

$$\text{CA}^{(\phi)}(t) = \frac{1}{|C_t|} \sum_{c \in C_t} \Pr(\hat{y}^\phi = c \mid y = c), \quad (5.1)$$

and report the macro-average $\frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \text{CA}(t)$ across evaluation datasets $\mathcal{T}$ so that no single large dataset dominates. For all detectors we use the default arg max decision on predicted class probabilities and do not tune thresholds per dataset. On real-only sets, CA corresponds to true-negative accuracy (1-FPR); on synthetic-only sets it corresponds to true-positive accuracy (1-FNR). We additionally compute AUC on the 24 mixed datasets (those containing both real and synthetic images) and report in Appendix Section F.

5.2. Closed-form Gaussian Discriminants versus Trained Heads

Tables 2 and 3 compares released AI-generated image detector heads with Gaussian discriminant rules fitted on the same frozen features and conditioned on the same training prior. Several rows in Tables 2 and 3 share the same underlying backbone. UnivFD and DRCT-UnivFD both use ViT-L-14-quickgelu-openai (Radford et al., 2021) as the image encoder; differences arise from the public training prior and the trained head. DRCT exposes classifier features after reconstruction training. Effort applies LoRA adaptation before its head. AIDE uses frequency/statistical experts plus semantic features, and CoDE/CF train backbones and heads end-to-end. All are auditable once the encoder’s final feature space is exposed.

Across several priors, the best closed-form Gaussian discriminant improves class accuracy, with the largest gains occurring when separation is well-captured by first- and second-order moments (often Mah-NCM). These patterns are consistent with Propositions 1 and 2: when separation is well-captured by first- and second-order moments in a frozen encoder feature space, the corresponding Gaussian

Table 4 | **Encoder scaling under a fixed prior: representation choice strongly affects transfer.** Mean class accuracy CA (macro-averaged across Percept-Lens evaluation datasets) for Gaussian discriminant heads conditioned on the same CommunityForensics training prior (Park and Owens, 2025), using frozen CLIP encoders of increasing capacity pretrained by OpenAI. Only the head is changed. No detector fine-tuning is performed.

Frozen Encoders	Euc-NCM (Wu et al., 2025b)	Cos-NCM	GNB	Mah-NCM	QDA
ResNet-50 (Timm, 2024a)	65.27%	64.36%	65.72%	72.00%	63.15%
ResNet-101 (He et al., 2016)	65.20%	64.70%	64.46%	70.27%	62.29%
ResNet-50x4 (Timm, 2024c)	66.24%	65.91%	65.12%	72.34%	64.28%
ResNet-50x16 (Timm, 2024b)	67.35%	66.77%	66.21%	74.98%	65.71%
ResNet-50x64 (Timm, 2024d)	68.42%	68.19%	65.02%	74.57%	68.38%
ViT-B/16 (Laion, 2024d)	64.66%	64.65%	66.37%	74.18%	65.70%
ViT-B/32 (Laion, 2024e)	64.53%	64.25%	64.83%	70.61%	62.81%
ViT-L/14 (Radford et al., 2021)	66.18%	68.44%	69.79%	76.96%	68.87%

Table 5 | **Pretraining objective affects transfer even without detector training.** Mean class accuracy CA (macro-averaged across Percept-Lens evaluation datasets) for Gaussian discriminant heads conditioned on the same CommunityForensics training prior (Park and Owens, 2025), using frozen encoders trained with different objectives.

Frozen Encoders	Euc-NCM (Wu et al., 2025b)	Cos-NCM	GNB	Mah-NCM	QDA
MAE-Huge (He et al., 2022)	55.18%	55.22%	58.33%	69.37%	64.01%
BEiT-Large (Bao et al., 2022)	58.53%	58.49%	58.80%	61.78%	53.63%
CLIP-XLM-RoBERTa-Large (Cherti et al., 2023)	70.72%	69.70%	70.61%	77.67%	65.79%
SigLIP-Large (Zhai et al., 2023)	58.25%	58.25%	58.07%	64.14%	60.85%
BLIP-Large (Li et al., 2022)	55.75%	56.89%	53.87%	70.17%	55.42%
BLIP2 (Li et al., 2023)	68.61%	68.46%	65.41%	77.73%	62.21%
DINOv2-giant (Oquab et al., 2024)	63.15%	63.59%	61.95%	73.30%	58.52%
DINOv3-ViT-7b (Siméoni et al., 2025)	83.85%	85.82%	83.18%	88.91%	78.88%

discriminants recover most of the available discriminative geometry. A main exception is CF-384 (Park and Owens, 2025), where the released detector remains ahead of the ladder (87.54% vs. 84.55% CA), consistent with a regime in which the trained head’s decision surface exploits structure not explained by second-order moments alone. Per-dataset results in Tables 14 and 21 also show large variability. For example, Mah-NCM reaches near-ceiling performance on many synthetic-only sets but is much weaker on DeepFakeBench (Yan et al., 2023), while QDA performs best on CelebA-Spoof (Zhang et al., 2020).

5.3. Encoder Sensitivity of Closed-form Gaussian Discriminants

We investigate the model-agnostic behavior of the ladder by evaluating a range of frozen architectures and pretraining objectives (Tables 4 and 5). Mah-NCM is consistently the top performer across these encoders, indicating that a shared full-covariance geometry captures a stable component of real/fake separability in modern representation spaces. This motivates a practical reporting guideline. Trained heads on frozen encoders should be compared against the best Gaussian rung on the *same* frozen features. Additional models are presented in Appendix Section E.

5.4. Training Prior Sensitivity of Closed-form Gaussian Discriminants

We isolate training-prior effects by fixing PE-Core-bigG-14-448 (Bolya et al., 2025) as the encoder and varying only the support prior used to estimate Gaussian parameters (Table 6). The spread is large: the best CA ranges from 77.92% (CNNSpot) to 94.46% (CommunityForensics), despite using the *same* representation and evaluation suite.

Moreover, the prevailing covariance assumption shifts with the prior: DRCT-2M favors QDA

Table 6 | **Training prior dominates transfer even under a fixed encoder.** Mean class accuracy CA (macro-averaged across Percept-Lens evaluation datasets) for Gaussian discriminant heads when varying only the *support prior* used to estimate moments, under the fixed frozen PE-Core-bigG-14-448 encoder (Bolya et al., 2025). Underlining denotes improvement over the best released head of CF-384 (Park and Owens, 2025) as reported in Table 2.

Training Dataset	Euc-NCM (Wu et al., 2025b)	Cos-NCM	GNB	Mah-NCM	QDA
CNNSpot (Wang et al., 2020)	74.95%	78.57%	59.21%	77.93%	49.24%
GenImage (Zhu et al., 2023)	86.68%	<u>88.75%</u>	87.50%	92.43%	81.83%
GenImage-SDv1	88.74%	<u>89.80%</u>	86.34%	92.57%	65.06%
DRCT-2M (Chen et al., 2024)	81.90%	81.76%	86.21%	83.25%	91.09%
DRCT-SDv1 (Chen et al., 2024)	82.29%	83.81%	<u>88.70%</u>	<u>88.31%</u>	89.92%
DRCT-SDv2 (Chen et al., 2024)	83.50%	84.38%	<u>89.56%</u>	<u>90.08%</u>	91.44%
ELSA-D3 (Baraldi et al., 2025)	<u>88.87%</u>	<u>89.03%</u>	<u>86.98%</u>	94.45%	86.40%
CommunityForensics (Park and Owens, 2025)	86.38%	86.06%	87.26%	94.46%	86.15%

Table 7 | **Data-efficiency of moment-based heads.** Mean class accuracy CA (macro-averaged across Percept-Lens evaluation datasets) for Gaussian discriminant heads as the support set used to estimate moments is subsampled from CommunityForensics (Park and Owens, 2025), under the fixed frozen PE-Core-bigG-14-448 encoder (Bolya et al., 2025). (Superscript/subscript denote max/min deviations (in percentage points) from the reported mean).

Amount	Euc-NCM (Wu et al., 2025b)	Cos-NCM	GNB	Mah-NCM	QDA
0.001%	83.97% ^{+1.93} _{-2.74}	83.46% ^{+2.25} _{-2.47}	87.14% ^{+1.21} _{-1.68}	85.56% ^{+1.84} _{-1.75}	85.13% ^{+1.07} _{-0.66}
0.005%	86.04% ^{+0.28} _{-0.40}	85.78% ^{+0.27} _{-0.53}	87.23% ^{+0.76} _{-0.77}	90.89% ^{+0.73} _{-1.34}	85.53% ^{+1.51} _{-1.71}
0.01%	86.32% ^{+0.67} _{-0.35}	86.02% ^{+0.77} _{-0.42}	87.42% ^{+1.19} _{-0.90}	91.55% ^{+0.67} _{-0.69}	86.48% ^{+1.16} _{-0.82}
0.1%	86.26% ^{+0.71} _{-0.44}	85.92% ^{+0.74} _{-0.47}	87.33% ^{+0.22} _{-0.19}	93.18% ^{+0.42} _{-0.54}	84.18% ^{+0.33} _{-0.71}
1%	86.39% ^{+0.11} _{-0.12}	86.05% ^{+0.11} _{-0.10}	87.28% ^{+0.04} _{-0.05}	93.52% ^{+0.28} _{-0.17}	85.16% ^{+0.20} _{-0.34}
100%	86.38%	86.06%	87.26%	94.46%	86.15%

(heteroscedastic) while all others favor Mah-NCM (homoscedastic). Thus, if two papers use the same backbone but different priors, their OOD behavior can differ more than what is attributable to the classifier family within the Gaussian ladder.

5.5. Data Efficiency of Closed-form Gaussian Discriminants

We investigate data efficiency by subsampling the support set used to estimate Gaussian parameters (Table 7). Performance saturates quickly. With only 1% of the prior (around 43k samples), Mah-NCM nearly matches the full-support estimate, while at extremely small support sizes (0.001% of the prior, or 44 samples) the diagonal GNB assumption can be more stable. The pattern reinforces the central observation from the ladder. For this support prior and encoder, low-order feature statistics capture a large share of the signal used by the best head, without iterative optimization.

With the frozen PE-Core-bigG-14-448 encoder (Bolya et al., 2025), Mah-NCM fitted from only 0.005% of the CommunityForensics prior (219 labeled samples) reaches 90.89% CA. Although this is not an apples-to-apples comparison to CF-384 (which uses a different backbone), it indicates that moment-based adaptation can be data-efficient once the representation is strong.

5.6. Representation Sensitivity of Wasserstein-2 Shift Estimates

All geometric shift metrics in Percept-Lens are computed in the encoder’s feature space, hence their numerical values are not purely properties of the data distributions, but of the *encoder* used to embed them. This raises a basic reliability question about whether conclusions drawn from a Gaussian shift metric are stable under reasonable encoder changes, or are artifacts of a particular backbone.

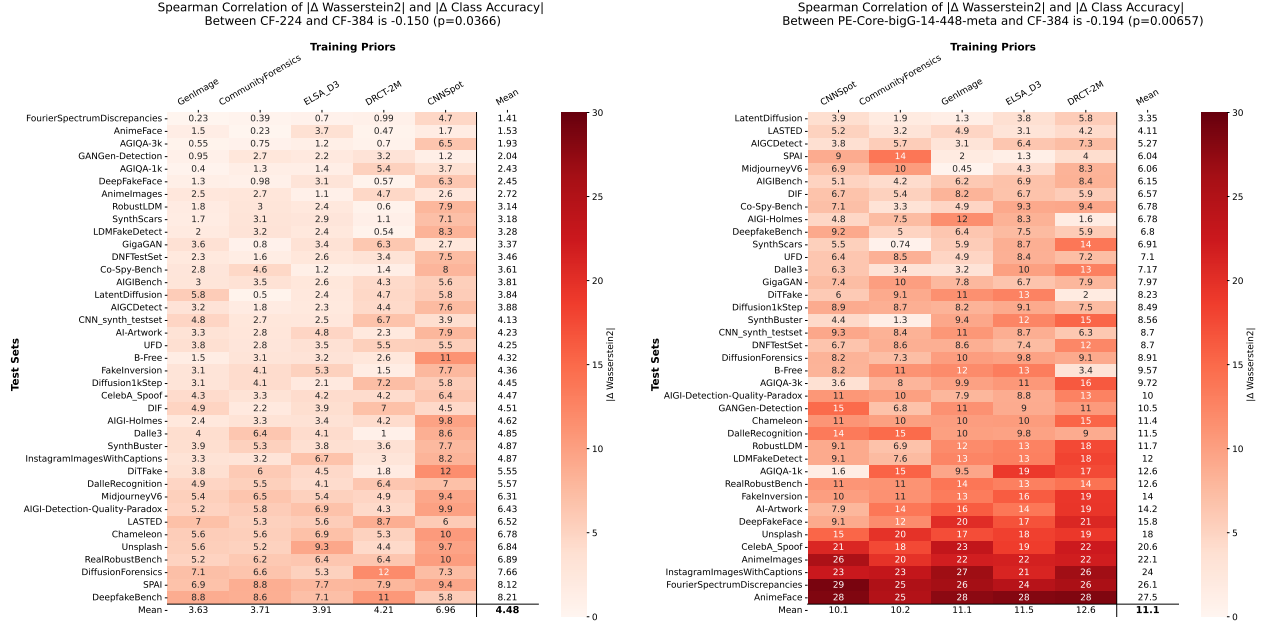


Figure 4 | **Gaussian Wasserstein-2 distances are representation-conditioned.** Each cell reports the absolute difference in class-averaged Gaussian $\mathcal{W}_2$ for the same train-test pair under two feature spaces, $\Delta \mathcal{W}_2(\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}}; \phi_a, \phi_b)$ (row/column means on the margins). The sub-figure title reports the Spearman correlation between the flattened cell-wise $\Delta \mathcal{W}_2$ values and the corresponding ΔCA values. Large changes in $\mathcal{W}_2$ do not necessarily coincide with large changes in Mah-NCM CA, so feature-space shift magnitudes should not be interpreted as detector-agnostic indicators without checking stability across representations.

Setup. For each training prior $\mathcal{D}_{\text{train}}$ and test set $\mathcal{D}_{\text{test}}$, class-conditional Gaussians are fitted to the representation induced by an encoder ϕ , then a class-averaged Wasserstein-2 shift is computed between the training and test distributions:

$$\mathcal{W}_2^{(\phi)}(\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}}) = \frac{1}{2} \sum_{c \in \{0,1\}} \mathcal{W}_2\left(\mathcal{N}(\mu_{c, \mathcal{D}_{\text{train}}}^{(\phi)}, \Sigma_{c, \mathcal{D}_{\text{train}}}^{(\phi)}), \mathcal{N}(\mu_{c, \mathcal{D}_{\text{test}}}^{(\phi)}, \Sigma_{c, \mathcal{D}_{\text{test}}}^{(\phi)})\right). \quad (5.2)$$

To isolate representation effects while keeping the data fixed, Figure 4 uses heatmap cells for the encoder-induced change in Gaussian $\mathcal{W}_2$ and panel titles for its Spearman correlation with the corresponding change in Mah-NCM accuracy. Thus each panel asks whether the train-test pairs whose estimated shift changes most after an encoder swap are also the pairs whose detector accuracy changes most. High correlation means the shift estimate is decision-aligned for that encoder pair. Weak correlation means $\mathcal{W}_2$ is responding to feature-space changes that the detector does not use:

$$\Delta \mathcal{W}_2(\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}}; \phi_a, \phi_b) = \left| \mathcal{W}_2^{(\phi_a)}(\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}}) - \mathcal{W}_2^{(\phi_b)}(\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}}) \right|, \quad (5.3)$$

$$\Delta \text{CA}(\mathcal{D}_{\text{test}}; \phi_a, \phi_b) = \left| \text{CA}^{(\phi_a)}(\mathcal{D}_{\text{test}}) - \text{CA}^{(\phi_b)}(\mathcal{D}_{\text{test}}) \right|. \quad (5.4)$$

Figure 4 shows strong representation dependence. Within a closely related encoder family, CF-224 and CF-384 (Park and Owens, 2025) have a modest $\Delta \mathcal{W}_2$ on average. Across larger backbone changes, such as PE-Core-bigG-14-448 (Bolya et al., 2025) versus CF-384 (Park and Owens, 2025), $\Delta \mathcal{W}_2$ increases by an order of magnitude, especially for real-only datasets. The sensitivity is also uneven.

Particular prior–test pairs exhibit consistently larger $\Delta\mathcal{W}_2$, indicating that some distribution shifts are more representation-fragile.

A weak association between $\Delta\mathcal{W}_2$ and ΔCA implies that global distributional distances can change under encoder re-parameterizations that do not materially affect the *discriminative* geometry used by Mah-NCM. For example, anisotropic rescaling can inflate Wasserstein distances while leaving the effective decision geometry largely intact. Percept-Lens therefore measures generalization at the level of complete detection systems (training prior $\times$ encoder $\times$ classifier), and the representation axis is not interchangeable. Any claim that uses $\mathcal{W}_2$ to argue for or against a training prior must therefore be stated as *representation-conditioned*. Robust conclusions should rely on patterns that persist across encoders (e.g., rank-consistent prior ordering or decision-aligned shift measures), rather than absolute $\mathcal{W}_2$ values in a single feature space.

Practical reporting checklist. These experiments suggest a minimal audit for future AIGI detector papers. Authors should report the released head and the best Gaussian rung on the same frozen features and support prior, include per-dataset CA/AUC rather than only macro averages, state the encoder used for any geometric shift metric, and flag non-Gaussian feature regimes using diagnostics such as those in Appendix Tables 23 and 24. This makes head-level OOD claims auditable without treating the Gaussian ladder as a replacement for specialized detectors.

6. Conclusion

This paper studies prior-conditioned Gaussian discriminants as practical baselines and diagnostics for AI-generated image detection under joint distribution shift. Using a unified public-data protocol, the ladder is often competitive with trained detector heads under matched priors and encoders, and sometimes exceeds them. We support a matched head audit plus controlled prior/encoder sweeps, not a fully symmetric decomposition of every detector component. The source-side recommendation is to compare the trained head with the best Gaussian rung on the same representation and support prior before claiming head-level OOD gains. Across priors, the winning covariance model indicates which low-order statistics transfer, and the data-efficiency results show that useful moment estimates can be obtained from small support sets. Future work should characterize failures in multimodal or heavy-tailed feature regimes and design adaptation procedures that improve OOD performance while preserving transferable geometry.

References

- Instagram images with captions. URL <https://www.kaggle.com/datasets/prithviijaunjal/instagram-images-with-captions>.
- terminusresearch/midjourney-v6-520k-raw · datasets at hugging face. URL <https://huggingface.co/datasets/terminusresearch/midjourney-v6-520k-raw>.
- Unsplash dataset. URL <https://unsplash.com/data>.
- Quentin Bammey. Synthbuster: Towards detection of diffusion model generated images. *IEEE Open Journal of Signal Processing*, 5:1–9, 2024. ISSN 2644-1322. doi: 10.1109/OJSP.2023.3337714. URL <https://ieeexplore.ieee.org/document/10334046>.
- Banana_Leopard. Anime images dataset, 2023. URL <https://www.kaggle.com/datasets/di-raizel/anime-images-dataset>.
- Parikshit Bansal, Ali Kavis, and Sujay Sanghavi. Understanding contrastive learning via gaussian mixture models. October 2025. URL <https://openreview.net/forum?id=nCAdkkAeR9>.

- Hangbo Bao, Li Dong, Songhao Piao, and Furu Wei. BEiT: BERT pre-training of image transformers. In *International Conference on Learning Representations*, 2022. URL <https://openreview.net/forum?id=p-BhZSz59o4>.
- Lorenzo Baraldi, Federico Cocchi, Marcella Cornia, Lorenzo Baraldi, Alessandro Nicolosi, and Rita Cucchiara. Contrasting deepfakes diffusion via contrastive learning and global-local similarities. In Aleš Leonardis, Elisa Ricci, Stefan Roth, Olga Russakovsky, Torsten Sattler, and Gül Varol, editors, *Computer Vision – ECCV 2024*, pages 199–216, Cham, 2025. Springer Nature Switzerland. ISBN 978-3-031-73036-8. doi: 10.1007/978-3-031-73036-8_12. URL <https://openreview.net/forum?id=tsrYdgxbFM>.
- Daniel Bolya, Po-Yao Huang, Peize Sun, Jang Hyun Cho, Andrea Madotto, Chen Wei, Tengyu Ma, Jiale Zhi, Jathushan Rajasegaran, Hanoona Rasheed, Junke Wang, Marco Monteiro, Hu Xu, Shiyu Dong, Nikhila Ravi, Daniel Li, Piotr Dollár, and Christoph Feichtenhofer. Perception encoder: The best visual embeddings are not at the output of the network, 2025. URL <https://arxiv.org/abs/2504.13181>.
- Brian Bushard. Fake image of explosion near pentagon went viral—even though it never happened. *Forbes*, 2023. URL <https://www.forbes.com/sites/brianbushard/2023/05/22/fake-image-of-explosion-near-pentagon-went-viral-even-though-it-never-happened/>.
- George Cazenavette, Avneesh Sud, Thomas Leung, and Ben Usman. FakeInversion: Learning to Detect Images from Unseen Text-to-Image Models by Inverting Stable Diffusion. pages 10759–10769, 2024. URL https://openaccess.thecvf.com/content/CVPR2024/html/Cazenavette_FakeInversion_Learning_to_Detect_Images_from_Unseen_Text-to-Image_Models_by_CVPR_2024_paper.html.
- Baoying Chen, Jishen Zeng, Jianquan Yang, and Rui Yang. DRCT: diffusion reconstruction contrastive training towards universal detection of diffusion generated images. In *Forty-first International Conference on Machine Learning*, June 2024. URL <https://openreview.net/forum?id=oRLwyayrh1>.
- Siyan Cheng, Lingjuan Lyu, Zhenting Wang, Xiangyu Zhang, and Vikash Sehwal. CO-SPY: Combining semantic and pixel features to detect synthetic images by AI. pages 13455–13465, 2025. URL https://openaccess.thecvf.com/content/CVPR2025/html/Cheng_CO-SPY_Combining_Semantic_and_Pixel_Features_to_Detect_Synthetic_Images_CVPR_2025_paper.html.
- Mehdi Cherti, Romain Beaumont, Ross Wightman, Mitchell Wortsman, Gabriel Ilharco, Cade Gordon, Christoph Schuhmann, Ludwig Schmidt, and Jenia Jitsev. Reproducible scaling laws for contrastive language-image learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2818–2829, 2023. URL https://openaccess.thecvf.com/content/CVPR2023/html/Cherti_Reproducible_Scaling_Laws_for_Contrastive_Language-Image_Learning_CVPR_2023_paper.html.
- Huan Liu Chuangchuang Tan, Renshuai Tao. GANGen-detection: a dataset generated by gans for generalizable deepfake detection, 2024. URL <https://github.com/chuangchuangtan/GANGen-Detection>.
- Spencer Churchill and Brian Chao. Anime face dataset, 2019. URL <https://www.kaggle.com/ds/379764>.
- Riccardo Corvi, Davide Cozzolino, Giada Zingarini, Giovanni Poggi, Koki Nagano, and Luisa Verdoliva. On the detection of synthetic images generated by diffusion models. In *ICASSP 2023 - 2023 IEEE*

- International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5, June 2023. doi: 10.1109/ICASSP49357.2023.10095167. URL <https://ieeexplore.ieee.org/abstract/document/10095167>.
- Lars Daniel. How hurricane helene deepfakes flooding social media hurt real people. *Forbes*, 2024. URL <https://www.forbes.com/sites/larsdaniel/2024/10/04/hurricane-helena-deepfakes-flooding-social-media-hurt-real-people/>.
- Tarik Dzanic, Karan Shah, and Freddie Witherden. Fourier spectrum discrepancies in deep network generated images. In *Advances in Neural Information Processing Systems*, volume 33, pages 3022–3032. Curran Associates, Inc., 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/1f8d87e1161af68b81bace188a1ec624-Abstract.html>.
- Ben Egan, Alex Redden, XWAVE, and SilentAntagonist. Dalle3 1 million+ high quality captions, May 2024. URL <https://huggingface.co/datasets/ProGamerGov/synthetic-dataset-1m-dalle3-high-quality-captions>.
- Joel Frank, Thorsten Eisenhofer, Lea Schönherr, Asja Fischer, Dorothea Kolossa, and Thorsten Holz. Leveraging Frequency Analysis for Deep Fake Image Recognition. In *Proceedings of the 37th International Conference on Machine Learning*, pages 3247–3258. PMLR, November 2020. URL <https://proceedings.mlr.press/v119/frank20a.html>.
- Fabrizio Guillaro, Giada Zingarini, Ben Usman, Avneesh Sud, Davide Cozzolino, and Luisa Verdoliva. A bias-free training paradigm for more general AI-generated image detection. pages 18685–18694, 2025. URL https://openaccess.thecvf.com/content/CVPR2025/html/Guillaro_A_Bias-Free_Training_Paradigm_for_More_General_AI-generated_Image_Detection_CVPR_2025_paper.html.
- Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 770–778, 2016. URL https://openaccess.thecvf.com/content_cvpr_2016/html/He_Deep_Residual_Learning_CVPR_2016_paper.html.
- Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked Autoencoders Are Scalable Vision Learners. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16000–16009, 2022. URL https://openaccess.thecvf.com/content/CVPR2022/html/He_Masked_Autoencoders_Are_Scalable_Vision_Learners_CVPR_2022_paper.html.
- Yinan He, Bei Gan, Siyu Chen, Yichun Zhou, Guojun Yin, Luchuan Song, Lu Sheng, Jing Shao, and Ziwei Liu. ForgeryNet: a versatile benchmark for comprehensive forgery analysis. pages 4360–4369, 2021. URL https://openaccess.thecvf.com/content/CVPR2021/html/He_ForgeryNet_A_Versatile_Benchmark_for_Comprehensive_Forgery_Analysis_CVPR_2021_paper.html.
- Gabriel Ilharco, Mitchell Wortsman, Ross Wightman, Cade Gordon, Nicholas Carlini, Rohan Taori, Achal Dave, Vaishaal Shankar, Hongseok Namkoong, John Miller, Hannaneh Hajishirzi, Ali Farhadi, and Ludwig Schmidt. OpenCLIP, July 2021. URL <https://doi.org/10.5281/zenodo.5143773>.
- Hengrui Kang, Siwei Wen, Zichen Wen, Junyan Ye, Weijia Li, Peilin Feng, Baichuan Zhou, Bin Wang, Dahua Lin, Linfeng Zhang, and Conghui He. LEGION: Learning to ground and explain for synthetic image detection. pages 18937–18947, 2025. URL https://openaccess.thecvf.com/content/ICCV2025/html/Kang_LEGION_Learning_to_Ground_and_Explain_for_Synthetic

[ic_Image_Detection_ICCV_2025_paper.html](#).

Minguk Kang, Jun-Yan Zhu, Richard Zhang, Jaesik Park, Eli Shechtman, Sylvain Paris, and Taesung Park. Scaling up GANs for text-to-image synthesis. pages 10124–10134, 2023. URL https://openaccess.thecvf.com/content/CVPR2023/html/Kang_Scaling_Up_GANs_for_Text-to-Image_Synthesis_CVPR_2023_paper.html.

Dimitrios Karageorgiou, Symeon Papadopoulos, Ioannis Kompatsiaris, and Efstratios Gavves. Any-Resolution AI-Generated Image Detection by Spectral Learning. pages 18706–18717, 2025. URL https://openaccess.thecvf.com/content/CVPR2025/html/Karageorgiou_Any-Resolution_AI-Generated_Image_Detection_by_Spectral_Learning_CVPR_2025_paper.html.

Adam El Kholy. AI-artwork, 2024. URL <https://www.kaggle.com/dsv/7878124>.

Nathan Koliha. AI recognition dataset, 2024. URL <https://www.kaggle.com/dsv/7501727>.

Laion. CLIP-convnext_base_w-laion2B-s13B-b82K-augreg, 2024a. URL https://huggingface.co/laion/CLIP-convnext_base_w-laion2B-s13B-b82K-augreg.

Laion. CLIP-convnext_large_d_320-laion2B-s29B-b131K-ft-soup, 2024b. URL https://huggingface.co/laion/CLIP-convnext_large_d_320-laion2B-s29B-b131K-ft-soup.

Laion. CLIP-convnext_xxlarge-laion2B-s34B-b82K-augreg-soup, 2024c. URL https://huggingface.co/laion/CLIP-convnext_xxlarge-laion2B-s34B-b82K-augreg-soup.

Laion. CLIP-ViT-B-16-datacomp.xl-s13B-b90K, 2024d. URL <https://huggingface.co/laion/CLIP-ViT-B-16-DataComp.XL-s13B-b90K>.

Laion. CLIP-ViT-B-32-256x256-DataComp-s34B-b86K, 2024e. URL <https://huggingface.co/laion/CLIP-ViT-B-32-256x256-DataComp-s34B-b86K>.

Kimin Lee, Kibok Lee, Honglak Lee, and Jinwoo Shin. A simple unified framework for detecting out-of-distribution samples and adversarial attacks. In *Advances in Neural Information Processing Systems*, volume 31. Curran Associates, Inc., 2018. URL https://papers.nips.cc/paper_files/paper/2018/hash/abdeb6f575ac5c6676b747bca8d09cc2-Abstract.html.

Chunxiao Li, Xiaoxiao Wang, Meiling Li, Boming Miao, Peng Sun, Yunjian Zhang, Xiangyang Ji, and Yao Zhu. Bridging the gap between ideal and real-world evaluation: Benchmarking AI-generated image detection in challenging scenarios. pages 20379–20389, 2025a. URL https://openaccess.thecvf.com/content/ICCV2025/html/Li_Bridging_the_Gap_Between_Ideal_and_Real-world_Evaluation_Benchmarking_AI-Generated_ICCV_2025_paper.html.

Chunyi Li, Zicheng Zhang, Haoning Wu, Wei Sun, Xiongkuo Min, Xiaohong Liu, Guangtao Zhai, and Weisi Lin. AGIQA-3K: an open database for AI-generated image quality assessment. *IEEE Transactions on Circuits and Systems for Video Technology*, 34(8):6833–6846, August 2024. ISSN 1558-2205. doi: 10.1109/TCSVT.2023.3319020. URL https://ieeexplore.ieee.org/abstract/document/10262331?casa_token=d0CSMRfrEDcAAAAA:hMig5SBv7x_oX--KIQDLr_fs5XcGZoKZr7VqsFMjujTKFwMNf32x5WilbDCrBp5l2Ht-REfDSw.

Junnan Li, Dongxu Li, Caiming Xiong, and Steven Hoi. BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 12888–12900. PMLR, 2022. URL <https://proceedings.mlr.press/v162/li22n.html>.

- Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 19730–19742. PMLR, 2023. URL <https://proceedings.mlr.press/v202/li23q.html>.
- Ouxiang Li, Jiayin Cai, Yanbin Hao, Xiaolong Jiang, Yao Hu, and Fuli Feng. Improving synthetic image detection towards generalization: an image transformation perspective. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.1*, KDD '25, pages 2405–2414, New York, NY, USA, July 2025b. Association for Computing Machinery. ISBN 979-8-4007-1245-6. doi: 10.1145/3690624.3709392. URL <https://dl.acm.org/doi/10.1145/3690624.3709392>.
- Ziqiang Li, Jiazhen Yan, Ziwen He, Kai Zeng, Weiwei Jiang, Lizhi Xiong, and Zhangjie Fu. Is artificial intelligence generated image detection a solved problem? October 2025c. URL <https://openreview.net/forum?id=N52U2h9k9o>.
- Bill McCarthy. Fake pentagon explosion image spreads online. *AFP Fact Check*, 2023. URL <https://factcheck.afp.com/doc.afp.com.33FV4BU>.
- MidJourney. MidJourney: AI-based text-to-image generation, 2022. URL <https://www.midjourney.com/>.
- Utkarsh Ojha, Yuheng Li, and Yong Jae Lee. Towards Universal Fake Image Detectors That Generalize Across Generative Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 24480–24489, 2023. URL https://openaccess.thecvf.com/content/CVPR2023/html/Ojha_Towards_Universal_Fake_Image_Detectors_That_Generalize_Across_Generative_Models_CVPR_2023_paper.html.
- Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jégou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024. URL <https://openreview.net/forum?id=a68SUt6zFt>.
- Jeongsoo Park and Andrew Owens. Community forensics: using thousands of generators to train fake image detectors. pages 8245–8257, 2025. URL https://openaccess.thecvf.com/content/CVPR2025/html/Park_Community_Forensics_Using_Thousands_of_Generators_to_Train_Fake_Image_CVPR_2025_paper.html.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 8748–8763. PMLR, 2021. URL <https://proceedings.mlr.press/v139/radford21a.html>.
- Anirudh Sundara Rajan and Yong Jae Lee. Stay-positive: a case for ignoring real image features in fake image detection. June 2025. URL <https://openreview.net/forum?id=VNLmfMJi3w¬eId=x7FR8R6HMU>.
- Anirudh Sundara Rajan, Utkarsh Ojha, Jedidiah Schloesser, and Yong Jae Lee. Aligned Datasets Improve Detection of Latent Diffusion-Generated Images. October 2024. URL <https://openreview.net/forum?id=doBkiqESYq>.

- Aditya Ramesh, Mikhail Pavlov, Gabriel Goh, Scott Gray, Chelsea Voss, Alec Radford, Mark Chen, and Ilya Sutskever. Zero-shot text-to-image generation. In *Proceedings of the 38th International Conference on Machine Learning*, pages 8821–8831. PMLR, July 2021. URL <https://proceedings.mlr.press/v139/ramesh21a.html>.
- Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10684–10695, 2022. URL https://openaccess.thecvf.com/content/CVPR2022/html/Rombach_High-Resolution_Image_Synthesis_With_Latent_Diffusion_Models_CVPR_2022_paper.
- Chitwan Saharia, William Chan, Saurabh Saxena, Lala Li, Jay Whang, Emily L. Denton, Kamyar Ghasemipour, Raphael Gontijo Lopes, Burcu Karagol Ayan, Tim Salimans, Jonathan Ho, David J. Fleet, and Mohammad Norouzi. Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding. *Advances in Neural Information Processing Systems*, 35:36479–36494, December 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/ec795aeadae0b7d230fa35cbaf04c041-Abstract-Conference.html.
- Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, Francisco Massa, Daniel Haziza, Luca Wehrstedt, Jianyuan Wang, Timothée Darcet, Théo Moutakanni, Leonel Sentana, Claire Roberts, Andrea Vedaldi, Jamie Tolan, John Brandt, Camille Couprie, Julien Mairal, Hervé Jégou, Patrick Labatut, and Piotr Bojanowski. DINOv3, 2025. URL <https://arxiv.org/abs/2508.10104>.
- Sergey Sinitisa and Ohad Fried. Deep image fingerprint: towards low budget synthetic image detection and model lineage analysis. pages 4067–4076, 2024. URL https://openaccess.thecvf.com/content/WACV2024/html/Sinitisa_Deep_Image_Fingerprint_Towards_Low_Budget_Synthetic_Image_Detection_and_WACV_2024_paper.html.
- Haixu Song, Shiyu Huang, Yinpeng Dong, and Wei-Wei Tu. Robustness and generalizability of deepfake detection: a study with diffusion models, September 2023. URL <http://arxiv.org/abs/2309.02218>.
- Chuangchuang Tan, Yao Zhao, Shikui Wei, Guanghua Gu, Ping Liu, and Yunchao Wei. Rethinking the up-sampling operations in CNN-based generative network for generalizable deepfake detection. pages 28130–28139, 2024. URL https://openaccess.thecvf.com/content/CVPR2024/html/Tan_Rethinking_the_Up-Sampling_Operations_in_CNN-based_Generative_Network_for_Generalizable_CVPR_2024_paper.html.
- Timm. Resnet50_clip.openai, 2024a. URL https://huggingface.co/timm/resnet50_clip.openai.
- Timm. Resnet50x16_clip.openai, 2024b. URL https://huggingface.co/timm/resnet50x16_clip.openai.
- Timm. Resnet50x4_clip.openai, 2024c. URL https://huggingface.co/timm/resnet50x4_clip.openai.
- Timm. Resnet50x64_clip.openai, 2024d. URL https://huggingface.co/timm/resnet50x64_clip.openai.
- Sheng-Yu Wang, Oliver Wang, Richard Zhang, Andrew Owens, and Alexei A. Efros. CNN-generated images are surprisingly easy to spot... for now. pages 8695–8704, 2020. URL https://openaccess.thecvf.com/content_CVPR_2020/html/Wang_CNN-Generated_Images_Are_Surp

[risingly_Easy_to_Spot..._for_Now_CVPR_2020_paper.html](#).

Zhendong Wang, Jianmin Bao, Wengang Zhou, Weilun Wang, Hezhen Hu, Hong Chen, and Houqiang Li. DIRE for Diffusion-Generated Image Detection. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 22388–22398. arXiv, October 2023. doi: 10.1109/ICCV51070.2023.02051. URL <https://ieeexplore.ieee.org/document/10377654/>.

Haiwei Wu, Jiantao Zhou, and Shile Zhang. Generalizable synthetic image detection via language-guided contrastive learning. *IEEE Transactions on Artificial Intelligence*, pages 1–11, 2025a. ISSN 2691-4581. doi: 10.1109/TAI.2025.3641104. URL <https://ieeexplore.ieee.org/abstract/document/11281880>.

Shiyu Wu, Jing Liu, Jing Li, and Yequan Wang. Few-Shot Learner Generalizes Across AI-Generated Image Detection. June 2025b. URL <https://openreview.net/forum?id=uvU29AfoNT¬eId=76cq12x3jK>.

Yao Xiao, Binbin Yang, Weiyan Chen, Jiahao Chen, Zijie Cao, ZiYi Dong, Xiangyang Ji, Liang Lin, Wei Ke, and Pengxu Wei. Are high-quality AI-generated images more difficult for models to detect? June 2025. URL <https://openreview.net/forum?id=sKYdVKE1tS¬eId=1BtdJodSwK>.

Shilin Yan, Ouxiang Li, Jiayin Cai, Yanbin Hao, Xiaolong Jiang, Yao Hu, and Weidi Xie. A sanity check for AI-generated image detection. October 2024. URL <https://openreview.net/forum?id=ODRHZrkOQM>.

Zhiyuan Yan, Yong Zhang, Xinhang Yuan, Siwei Lyu, and Baoyuan Wu. DeepfakeBench: A comprehensive benchmark of deepfake detection. *Advances in Neural Information Processing Systems*, 36:4534–4565, December 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/0e735e4b4f07de483cbe250130992726-Abstract-Datasets_and_Benchmarks.html.

Zhiyuan Yan, Jiangming Wang, Peng Jin, Ke-Yue Zhang, Chengchun Liu, Shen Chen, Taiping Yao, Shouhong Ding, Baoyuan Wu, and Li Yuan. Orthogonal subspace decomposition for generalizable AI-generated image detection. June 2025. URL <https://openreview.net/forum?id=GFpj08S8Po>.

Muli Yang, Gabriel James Goenawan, Henan Wang, Huaiyuan Qin, Chenghao Xu, Yanhua Yang, Fen Fang, Ying Sun, Joo Hwee Lim, and Hongyuan Zhu. Your AI-Generated Image Detector Can Secretly Achieve SOTA Accuracy, If Calibrated. *Proceedings of the AAAI Conference on Artificial Intelligence*, 40(14):11622–11630, March 2026. ISSN 2374-3468. doi: 10.1609/aaai.v40i14.38146. URL <https://ojs.aaai.org/index.php/AAAI/article/view/38146>.

Xin Yang, Yuezun Li, and Siwei Lyu. Exposing Deep Fakes Using Inconsistent Head Poses. In *ICASSP 2019 - 2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 8261–8265, May 2019. doi: 10.1109/ICASSP.2019.8683164. URL https://ieeexplore.ieee.org/abstract/document/8683164?casa_token=J-Gs5Rxe4HEAAAAA:67CcBzvyD6o9fsp18ns9tw6bsA4VIADNo5cnBcwvxYl3vHgtPqWCs8k8pWb0eWNRQRO-HWAg2g.

Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. Sigmoid loss for language image pre-training. In *IEEE/CVF International Conference on Computer Vision*, pages 11975–11986, 2023. URL https://openaccess.thecvf.com/content/ICCV2023/html/Zhai_Sigmoid_Loss_for_Language_Image_Pre-Training_ICCV_2023_paper.html.

Yichi Zhang and Xiaogang Xu. Diffusion noise feature: Accurate and fast generated image detection. In *Ecai 2025*, pages 1139–1146. IOS Press, 2025. doi: 10.3233/FAIA250925. URL <https://ebooks.iospress.nl/doi/10.3233/FAIA250925>.

- Yuanhan Zhang, ZhenFei Yin, Yidong Li, Guojun Yin, Junjie Yan, Jing Shao, and Ziwei Liu. CelebA-spoof: Large-scale face anti-spoofing dataset with rich annotations. In Andrea Vedaldi, Horst Bischof, Thomas Brox, and Jan-Michael Frahm, editors, *Computer Vision – ECCV 2020*, pages 70–85, Cham, 2020. Springer International Publishing. ISBN 978-3-030-58610-2. doi: 10.1007/978-3-030-58610-2_5.
- Zicheng Zhang, Chunyi Li, Wei Sun, Xiaohong Liu, Xiongkuo Min, and Guangtao Zhai. A perceptual quality assessment exploration for AIGC images. In *2023 IEEE International Conference on Multimedia and Expo Workshops (ICMEW)*, pages 440–445, July 2023. doi: 10.1109/ICMEW59549.2023.00082. URL <https://ieeexplore.ieee.org/document/10222021>.
- Nan Zhong, Yiran Xu, Sheng Li, Zhenxing Qian, and Xinpeng Zhang. PatchCraft: Exploring texture patch for efficient AI-generated image detection, March 2024. URL <http://arxiv.org/abs/2311.12397>.
- Yue Zhou, Xinan He, Kaiqing Lin, Bing Fan, Feng Ding, Jinhua Zeng, and Bin Li. Brought a Gun to a Knife Fight: Modern VFM Baselines Outgun Specialized Detectors on In-the-Wild AI Image Detection, October 2025a. URL <http://arxiv.org/abs/2509.12995>.
- Ziyin Zhou, Yunpeng Luo, Yuanchen Wu, Ke Sun, Jiayi Ji, Ke Yan, Shouhong Ding, Xiaoshuai Sun, Yunsheng Wu, and Rongrong Ji. AIGI-holmes: Towards explainable and generalizable AI-generated image detection via multimodal large language models. pages 18746–18758, 2025b. URL https://openaccess.thecvf.com/content/ICCV2025/html/Zhou_AIGI-Holmes_Towards_Explainable_and_Generalizable_AI-Generated_Image_Detection_via_Multimodal_ICCV_2025_paper.html.
- Mingjian Zhu, Hanting Chen, Qiangyu Yan, Xudong Huang, Guanyu Lin, Wei Li, Zhijun Tu, Hailin Hu, Jie Hu, and Yunhe Wang. GenImage: A Million-Scale Benchmark for Detecting AI-Generated Image. *Advances in Neural Information Processing Systems*, 36:77771–77782, December 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/f4d4a021f9051a6c18183b059117e8b5-Abstract-Datasets_and_Benchmarks.html.

A. On the Computational Paradigm and Practical Utility of Gaussian Discriminants

Closed-form Gaussian heads differ from gradient-trained heads not only in statistical assumptions but also in optimization burden. Fitting a rung of the ladder reduces to estimating sample means and (possibly regularized) covariances on a support set, followed by a single solve. This procedure is deterministic and largely hyperparameter-light. Aside from standard covariance regularization (e.g., shrinkage or diagonal loading), there is no learning-rate schedule, early stopping, or multi-run tuning.

The computational cost depends on the covariance structure. Isotropic and diagonal heads require only per-dimension statistics ($O(ND)$). Full-covariance heads (Mah-NCM/QDA) additionally estimate a $D \times D$ covariance ($O(ND^2)$) and invert or solve linear systems ($O(D^3)$), which can be non-trivial for very large D or very large N . However, this cost is paid once per (prior, encoder) pair and does not scale with the number of training epochs.

In contrast, gradient-based heads typically require many passes over the data and introduce additional variance from stochastic optimization and hyperparameter choices. For our purposes, this simplicity is a feature. The Gaussian ladder provides a controlled baseline and diagnostic, while its one-shot fitting reduces optimization confounds and makes head comparisons easier to reproduce.

B. Proofs of Theoretical Propositions

We provide formal derivations for the propositions presented in the theoretical framework Section 4.1. We adopt the notation defined in the main text.

B.1. Proof of Proposition 1 (Bayes-Optimal Classifier under Gaussian Assumption)

Proposition 5. *If the class-conditional densities are Gaussian, $p(z | y = c) \sim \mathcal{N}(\mu_c, \Sigma_c)$, and the class priors are equal, $P(y = 0) = P(y = 1) = \frac{1}{2}$, the decision rule that minimizes the probability of error (the Bayes-optimal rule) is given by Quadratic Discriminant Analysis (QDA).*

Proof. The Bayes-optimal decision rule minimizes the probability of misclassification by assigning a feature vector z to the class c with the maximum a posteriori (MAP) probability, $P(y = c|z)$. For a binary classification task, this means that we assign z to class 1 if $P(y = 1|z) > P(y = 0|z)$ and to class 0 otherwise.

Formulating the Decision Rule. Using Bayes' theorem, the posterior probability is $P(y = c|z) = \frac{p(z|y=c)P(y=c)}{p(z)}$. The MAP decision rule is therefore:

$$\frac{p(z|y=1)P(y=1)}{p(z)} > \frac{p(z|y=0)P(y=0)}{p(z)} \quad (\text{B.1})$$

Since the evidence $p(z)$ is a positive common denominator and we assumed that the class priors are equal ($P(y=1) = P(y=0) = \frac{1}{2}$). The decision rule simplifies to a comparison of the class-conditional likelihoods:

$$p(z|y=1) > p(z|y=0) \quad (\text{B.2})$$

We define the decision function $g^*(z)$ as the log-likelihood ratio:

$$g^*(z) = \log p(z|y=1) - \log p(z|y=0) \quad (\text{B.3})$$

The decision rule using the log-likelihood ratio is to classify as class 1 if $g^*(z) > 0$ and class 0 otherwise.

Introducing Gaussian PDF assumption to decision rule. The probability density function (PDF) for a multivariate Gaussian distribution is:

$$p(z|y=c) = \mathcal{N}(z; \mu_c, \Sigma_c) = \frac{1}{(2\pi)^{D/2} |\Sigma_c|^{1/2}} \exp\left(-\frac{1}{2}(z - \mu_c)^\top \Sigma_c^{-1} (z - \mu_c)\right) \quad (\text{B.4})$$

The corresponding log-likelihood for class c is:

$$\log p(z|y=c) = -\frac{D}{2} \log(2\pi) - \frac{1}{2} \log |\Sigma_c| - \frac{1}{2} (z - \mu_c)^\top \Sigma_c^{-1} (z - \mu_c) \quad (\text{B.5})$$

Using this expression in the decision function $g^*(z)$ Equation (B.3), we convert our decision rule into an approximation using Gaussian parameters after simplification.

$$g^*(z) \propto (z - \mu_0)^\top \Sigma_0^{-1} (z - \mu_0) - (z - \mu_1)^\top \Sigma_1^{-1} (z - \mu_1) + \log \frac{|\Sigma_0|}{|\Sigma_1|} \quad (\text{B.6})$$

This is the discriminant function for Quadratic Discriminant Analysis (QDA), which is therefore Bayes-optimal under the stated Gaussian assumption. $\square$

B.2. Proof of Proposition 2 (Optimality under Homoscedasticity)

Proposition 6. *If the classes are additionally assumed to be homoscedastic (i.e., they share a common covariance matrix $\Sigma_0 = \Sigma_1 = \Sigma_p$), the decision rule that minimizes the probability of error (the Bayes-optimal rule) is given by Mahalanobis Nearest Centroid Matching (Mah-NCM).*

Proof. We begin with the Bayes-optimal QDA decision function derived above Equation (B.6) and apply the homoscedasticity assumption $\Sigma_0 = \Sigma_1 = \Sigma_p$ to the decision function Equation (B.6) and after simplifications of the quadratic forms, the decision rule becomes of the form:

$$g^*(z) \propto (z - \mu_0)^\top \Sigma_p^{-1} (z - \mu_0) - (z - \mu_1)^\top \Sigma_p^{-1} (z - \mu_1) \quad (\text{B.7})$$

$$\propto \left(z^\top \Sigma_p^{-1} z - 2\mu_0^\top \Sigma_p^{-1} z + \mu_0^\top \Sigma_p^{-1} \mu_0 \right) - \left(z^\top \Sigma_p^{-1} z - 2\mu_1^\top \Sigma_p^{-1} z + \mu_1^\top \Sigma_p^{-1} \mu_1 \right) \quad (\text{B.8})$$

$$\propto 2\mu_1^\top \Sigma_p^{-1} z - 2\mu_0^\top \Sigma_p^{-1} z + \mu_0^\top \Sigma_p^{-1} \mu_0 - \mu_1^\top \Sigma_p^{-1} \mu_1 \quad (\text{B.9})$$

$$\propto 2(\mu_1 - \mu_0)^\top \Sigma_p^{-1} z - (\mu_1^\top \Sigma_p^{-1} \mu_1 - \mu_0^\top \Sigma_p^{-1} \mu_0) \quad (\text{B.10})$$

$$\propto 2(\mu_1 - \mu_0)^\top \Sigma_p^{-1} z - (\mu_1 - \mu_0)^\top \Sigma_p^{-1} (\mu_1 + \mu_0) \quad (\text{B.11})$$

Note that this assumption of homoscedasticity $\Sigma_0 = \Sigma_1 = \Sigma_p$ linearizes the decision boundary:

$$g^*(z) \propto w^\top z + b \quad (\text{B.12})$$

$$\text{where } w = \Sigma_p^{-1} (\mu_1 - \mu_0) \quad (\text{B.13})$$

$$b = -\frac{1}{2} (\mu_1 + \mu_0)^\top w = -\frac{1}{2} (\mu_1 + \mu_0)^\top \Sigma_p^{-1} (\mu_1 - \mu_0) \quad (\text{B.14})$$

This is the discriminant function for Mahalanobis Nearest Centroid Matching (Mah-NCM), which is therefore Bayes-optimal under the stated homoscedastic Gaussian assumption. $\square$

B.3. Proof Sketch for Proposition 3 (Performance Stability under Bounded Drift)

Proposition 7. Assume a homoscedastic classifier with shared covariance Σ and equal priors. If the statistical drift between the training and testing distributions is bounded by $\|\mu_c^{\text{test}} - \mu_c^{\text{train}}\|_2 \leq \epsilon_\mu$ and $\|\Sigma^{\text{test}} - \Sigma^{\text{train}}\|_F \leq \epsilon_\Sigma$, then $|\text{AUC}_{\text{test}} - \text{AUC}_{\text{train}}| \leq L(\epsilon_\mu + \epsilon_\Sigma)$.

Proof Sketch. A full proof requires extensive details from perturbation theory. We provide a rigorous sketch outlining the main logical steps.

Expressing AUC as a function of statistical parameters. For a linear classifier with weight vector w , the AUC is the probability that a randomly drawn positive sample scores higher than a randomly drawn negative sample: $\text{AUC} = P(w^\top Z_1 > w^\top Z_0)$, where $Z_c \sim \mathcal{N}(\mu_c, \Sigma)$. Let the score difference be the random variable $Y = w^\top (Z_1 - Z_0)$. Since Z_1 and Z_0 are independent Gaussian variables, their difference is also Gaussian: $Z_1 - Z_0 \sim \mathcal{N}(\mu_1 - \mu_0, 2\Sigma)$. Therefore, Y is a scalar Gaussian with mean $\mathbb{E}[Y] = w^\top (\mu_1 - \mu_0)$ and variance $\text{Var}(Y) = w^\top (2\Sigma)w$. The AUC is $P(Y > 0)$, which can be expressed using the CDF of the standard normal distribution, Φ :

$$\text{AUC} = P\left(\frac{Y - \mathbb{E}[Y]}{\sqrt{\text{Var}(Y)}} > \frac{-\mathbb{E}[Y]}{\sqrt{\text{Var}(Y)}}\right) = \Phi\left(\frac{\mathbb{E}[Y]}{\sqrt{\text{Var}(Y)}}\right) \quad (\text{B.15})$$

Under the homoscedastic Gaussian assumption, for the optimal linear classifier Mahalanobis Nearest Centroid Matching (Mah-NCM) where $w = \Sigma^{-1}(\mu_1 - \mu_0)$, this simplifies to:

$$\text{AUC} = \Phi\left(\frac{(\mu_1 - \mu_0)^\top \Sigma^{-1}(\mu_1 - \mu_0)}{\sqrt{2(\mu_1 - \mu_0)^\top \Sigma^{-1} \Sigma \Sigma^{-1}(\mu_1 - \mu_0)}}\right) = \Phi\left(\sqrt{\frac{\mathcal{D}_\mu}{2}}\right) \quad (\text{B.16})$$

This establishes that the AUC is a direct and smooth function of the Fisher margin $\mathcal{D}_\mu$.

Establishing Lipschitz Continuity. It remains to show that AUC is locally Lipschitz. The map is a composition of three parts:

1. The Fisher Margin $\mathcal{D}_\mu = (\mu_1 - \mu_0)^\top \Sigma^{-1}(\mu_1 - \mu_0)$ is a smooth (infinitely differentiable) function of its arguments as long as Σ is invertible. (Note that all the training datasets we used in the experiments made this assumption of invertible Σ valid by ensuring $n \gg d$, future works can explore the shrinkage Σ using Ledoit-Wolf Shrinkage when training sets are of form $n \ll d$, ensuring Σ is invertible).
2. The square root function, which is locally Lipschitz in $(0, \infty)$.
3. The standard normal CDF $\Phi(x)$, which is globally Lipschitz because its derivative (the normal PDF) is bounded by $1/\sqrt{2\pi}$.

Since the composition of local Lipschitz functions is locally Lipschitz, the overall function AUC is locally Lipschitz.

Deriving the bounds on deviation. By the definition of local Lipschitz continuity, for small perturbations in the arguments, the change in the function's value is bounded by a constant times the magnitude of the perturbation. Therefore, for perturbations bounded by ϵ_μ and ϵ_Σ , we have

$$|\text{AUC}_{\text{test}} - \text{AUC}_{\text{train}}| = |H(\mu^{\text{test}}, \Sigma^{\text{test}}) - H(\mu^{\text{train}}, \Sigma^{\text{train}})| \leq L \cdot (\|\mu^{\text{test}} - \mu^{\text{train}}\| + \|\Sigma^{\text{test}} - \Sigma^{\text{train}}\|) \quad (\text{B.17})$$

This can be expressed as $|\text{AUC}_{\text{test}} - \text{AUC}_{\text{train}}| \leq L_1 \epsilon_\mu + L_2 \epsilon_\Sigma \leq L(\epsilon_\mu + \epsilon_\Sigma)$, where $L = \max(L_1, L_2)$. The Lipschitz constant L depends on the local derivatives of H , which are functions of the training parameters, critically including the spectral norm of the inverse covariance, $\|\Sigma^{-1}\|_2$. $\square$

B.4. Proof of Proposition 4 (Stability of the Fisher Margin under Distributional Drift)

Proposition 8. Assume that both the training and testing domains are characterized by homoscedastic Gaussian parameters $(\mu_c^{\text{train}}, \Sigma_p^{\text{train}})$ and $(\mu_c^{\text{test}}, \Sigma_p^{\text{test}})$, respectively. If the distributional drift is bounded such that $\|\delta^{\text{test}} - \delta^{\text{train}}\|_2 \leq \epsilon$ and $\|\Sigma_p^{\text{test}} - \Sigma_p^{\text{train}}\|_F \leq \eta$, where $\delta = \mu_1 - \mu_0$, then the absolute change in the Fisher Margin $\mathcal{D}_\mu$ is bounded, to a first order, by $|\mathcal{D}_\mu^{\text{test}} - \mathcal{D}_\mu^{\text{train}}| \leq 2\|(\Sigma_p^{\text{train}})^{-1}\|_2\|\delta^{\text{train}}\|_2\epsilon + O(\eta)$.

Proof. We analyze the change in Fisher Margin $\mathcal{D}_\mu = \delta^\top \Sigma_p^{-1} \delta$ under small perturbations. We define perturbations as $\Delta\delta = \delta^{\text{test}} - \delta^{\text{train}}$ and $\Delta\Sigma_p = \Sigma_p^{\text{test}} - \Sigma_p^{\text{train}}$, with norms bounded by $\|\Delta\delta\|_2 \leq \epsilon$ and $\|\Delta\Sigma_p\|_F \leq \eta$.

First-Order Taylor Expansion. On expanding $\mathcal{D}_\mu$ around the training parameters $(\delta^{\text{train}}, \Sigma_p^{\text{train}})$, in first order, the change is given by the total derivative:

$$\Delta\mathcal{D}_\mu = \mathcal{D}_\mu^{\text{test}} - \mathcal{D}_\mu^{\text{train}} \approx (\nabla_\delta \mathcal{D}_\mu)^\top \Delta\delta + \langle \nabla_{\Sigma_p} \mathcal{D}_\mu, \Delta\Sigma_p \rangle_F \quad (\text{B.18})$$

where the gradients are evaluated at the training parameters.

Gradient with respect to δ .

$$\nabla_\delta \mathcal{D}_\mu = 2\Sigma_p^{-1} \delta \quad (\text{B.19})$$

Gradient with respect to Σ_p .

$$d\mathcal{D}_\mu = (d\delta^\top) \Sigma_p^{-1} \delta + \delta^\top \Sigma_p^{-1} (d\delta) + \delta^\top (d\Sigma_p^{-1}) \delta \quad (\text{B.20})$$

$$= 2\delta^\top \Sigma_p^{-1} d\delta - \delta^\top \Sigma_p^{-1} (d\Sigma_p) \Sigma_p^{-1} \delta \quad (\text{B.21})$$

$$= \text{Tr} \left(2\delta^\top \Sigma_p^{-1} d\delta \right) - \text{Tr} \left(\delta^\top \Sigma_p^{-1} (d\Sigma_p) \Sigma_p^{-1} \delta \right) \quad (\text{B.22})$$

$$= \text{Tr} \left(2\delta^\top \Sigma_p^{-1} d\delta \right) - \text{Tr} \left(\Sigma_p^{-1} \delta \delta^\top \Sigma_p^{-1} d\Sigma_p \right) \quad (\text{B.23})$$

From the definition of the Frobenius inner product, $\langle A, B \rangle_F = \text{Tr}(A^\top B)$, the gradient with respect to Σ_p is:

$$\nabla_{\Sigma_p} \mathcal{D}_\mu = - \left(\Sigma_p^{-1} \delta \delta^\top \Sigma_p^{-1} \right)^\top = -\Sigma_p^{-1} \delta \delta^\top \Sigma_p^{-1} \quad (\text{B.24})$$

since Σ_p is symmetric.

Deriving the bound on the magnitude of the change. Using the triangle inequality on the first-order expansion Equation (B.18):

$$|\Delta\mathcal{D}_\mu| \leq \left| (\nabla_\delta \mathcal{D}_\mu)^\top \Delta\delta \right| + \left| \langle \nabla_{\Sigma_p} \mathcal{D}_\mu, \Delta\Sigma_p \rangle_F \right| \quad (\text{B.25})$$

We bound each term separately:

Mean Drift Term. By the Cauchy-Schwarz inequality:

$$\begin{aligned} \left| \left(2(\Sigma_p^{\text{train}})^{-1} \delta^{\text{train}} \right)^\top \Delta\delta \right| &\leq \|2(\Sigma_p^{\text{train}})^{-1} \delta^{\text{train}}\|_2 \|\Delta\delta\|_2 \\ &\leq 2\|(\Sigma_p^{\text{train}})^{-1}\|_2 \|\delta^{\text{train}}\|_2 \epsilon \end{aligned}$$

where $\|\cdot\|_2$ is the spectral norm for matrices and the Euclidean norm for vectors.

Covariance Drift Term. Using the property of the Frobenius inner product, $|\langle A, B \rangle_F| \leq \|A\|_F \|B\|_F$:

$$\begin{aligned} |\langle \nabla_{\Sigma_p} \mathcal{D}_\mu, \Delta \Sigma_p \rangle_F| &\leq \|\nabla_{\Sigma_p} \mathcal{D}_\mu\|_F \|\Delta \Sigma_p\|_F \\ &\leq C \cdot \eta = O(\eta) \end{aligned}$$

where $C = \|(\Sigma_p^{\text{train}})^{-1} \delta^{\text{train}} (\delta^{\text{train}})^\top (\Sigma_p^{\text{train}})^{-1}\|_F$ is a constant determined by the training distribution.

Combining the bounds for both terms yields:

$$\left| \mathcal{D}_\mu^{\text{test}} - \mathcal{D}_\mu^{\text{train}} \right| \leq 2 \|(\Sigma_p^{\text{train}})^{-1}\|_2 \|\delta^{\text{train}}\|_2 \epsilon + O(\eta) \quad (\text{B.26})$$

This shows that the change in the Fisher Margin $\mathcal{D}_\mu$ is controlled, to a first order, by the magnitude of the drift in the class statistics. $\square$

C. Analytical Forms of the Gaussian Discriminant Ladder

For each rung, we estimate class means μ_c and (optionally) covariance matrices from a labeled *support* set in the training prior. Given a test feature z , the heads produce unnormalized class scores $g_c(z)$ and predict $\hat{y}(z) = \arg \max_{c \in \{0,1\}} g_c(z)$. The score functions used in Table 1 are listed below. (Additive constants shared across classes are omitted.)

Class priors. To avoid confounding head comparisons with class-imbalance in the support prior, we use *uniform* class priors by default ($\pi_0 = \pi_1 = \frac{1}{2}$), so any $\log \pi_c$ terms are class-independent and may be dropped. When we report results with empirical priors, π_c denotes the support-set class frequency and is stated explicitly. Balanced evaluation metrics (e.g., balanced accuracy) do *not* in general remove prior-induced offsets in the decision rule.

Euc-NCM (isotropic, shared covariance).

$$g_c(z) = -\|z - \mu_c\|_2^2. \quad (\text{C.1})$$

Cos-NCM (cosine similarity).

$$g_c(z) = \left\langle \frac{z}{\|z\|_2}, \frac{\mu_c}{\|\mu_c\|_2} \right\rangle. \quad (\text{C.2})$$

GNB (diagonal covariance). Let $\Sigma_c = \text{diag}(\sigma_c^2)$.

$$g_c(z) = -\frac{1}{2} \sum_{j=1}^d \left(\frac{(z_j - \mu_{c,j})^2}{\sigma_{c,j}^2} + \log \sigma_{c,j}^2 \right) + \log \pi_c. \quad (\text{C.3})$$

Mah-NCM / LDA (shared full covariance). Let $\Sigma_0 = \Sigma_1 = \Sigma$.

$$g_c(z) = z^\top \Sigma^{-1} \mu_c - \frac{1}{2} \mu_c^\top \Sigma^{-1} \mu_c + \log \pi_c. \quad (\text{C.4})$$

QDA (class-specific full covariance).

$$g_c(z) = -\frac{1}{2} (z - \mu_c)^\top \Sigma_c^{-1} (z - \mu_c) - \frac{1}{2} \log \det(\Sigma_c) + \log \pi_c. \quad (\text{C.5})$$

(With the default uniform priors used in this paper, $\log \pi_c$ is constant across c and is omitted in implementation.)

Table 8 | **Summary of training priors used for conditioning.** We list the public datasets used as training priors in our experiments and report the number of real and synthetic images available in the version used by our pipeline. ♣ indicates that counts correspond to the publicly available subset used in our preprocessing and may differ from the full dataset described in the original source due to corrupted images in the original source.

Training Dataset	Total Images	Real	Synthetic
CNNSpot (Wang et al., 2020)	720,119	360,059	360,060
GenImage ♣ (Zhu et al., 2023)	2,254,762	1,116,779	1,137,983
GenImage-SDv1 (Zhu et al., 2023)	323,997	162,000	161,997
DRCT-2M ♣ (Chen et al., 2024)	2,247,453	118,287	2,129,166
DRCT-SDv1 (Chen et al., 2024)	473,148	118,287	354,861
DRCT-SDv2 (Chen et al., 2024)	473,148	118,287	354,861
CommunityForensics ♣ (Park and Owens, 2025)	4,386,820	1,678,386	2,708,434
ELSA-D3 ♣ (Baraldi et al., 2025)	8,417,550	1,683,511	6,734,039

D. Details about existing datasets in our Percept-Lens evaluation suite

The following tables list the datasets used for training (Table 8) and evaluation (Table 9) to make the experimental setup reproducible. Percept-Lens uses public datasets, leaves support priors out of evaluation, and combines generator style, source-domain, and post-processing shifts rather than a single generator-only shift.

Covariance estimation and numerical stability. For full-covariance rules (Mah-NCM/QDA), covariance inversion in high-dimensional feature spaces can be ill-conditioned, especially in low-shot regimes. We therefore regularize covariance estimates before computing Σ^{-1} and $\log \det(\Sigma)$. Concretely, we use an empirical covariance estimate with diagonal loading, $\hat{\Sigma} \leftarrow \hat{\Sigma} + \epsilon I$, and compute inverses via a numerically stable pseudo-inverse (small singular values truncated) to avoid catastrophic failures when $\hat{\Sigma}$ is nearly singular. Log-determinants are computed from the (clipped) spectrum of $\hat{\Sigma}$ to guarantee finite values. All priors and encoders are evaluated with the *same* estimator and regularization hyperparameters to avoid confounding.

We use standard regularization (shrinkage estimators such as Ledoit-Wolf, plus diagonal loading ϵI) before computing Σ^{-1} and log-determinants, when computing the covariances for data-efficiency results in Tables 7 and 22. This is because under extreme data shrinkage, we enter the $N < D$ regime (number of samples smaller than feature dimension), where empirical covariance estimates are unstable. All priors and backbones are evaluated with the same estimator to avoid confounding.

Table 9 | **Summary of Percept-Lens evaluation datasets.** We list the public datasets used to evaluate generalization and report the number of real and synthetic images. ♦ indicates that we evaluate on all officially released subsets/variants provided by the source; when a dataset provides multiple transformations of the *same* underlying image, we treat each transformation as a separate evaluation subset and do not interpret the resulting counts as independent samples.

Evaluation Dataset	Total Images	Real	Synthetic
Out-of-Distribution Existing Datasets with only Real Images			
Unsplash (Lite Subset) (uns)	24,969	24,969	—
InstagramImagesWithCaptions (ins)	34,927	34,927	—
Anime Faces Dataset (Churchill and Chao, 2019)	63,565	63,565	—
Anime Images (Banana_Leopard, 2023)	82,975	82,975	—
CelebA-Spoof ♦ (Zhang et al., 2020)	561,575	561,575	—

Continued on next page

Evaluation Dataset	Total Images	Real	Synthetic
Out-of-Distribution Existing Datasets with only Synthetic Images			
AGIQA-1k (Zhang et al., 2023)	1,080	—	1,080
AGIQA-3k (Li et al., 2024)	2,982	—	2,982
SPAI (Karageorgiou et al., 2025)	3,638	—	3,638
SynthBuster Extended (Bammey, 2024; Guillaro et al., 2025)	11,003	—	11,003
SynthScars (Kang et al., 2025)	12,182	—	12,182
GigaGAN (Kang et al., 2023)	170,000	—	170,000
LatentDiffusion (Corvi et al., 2023)	216,000	—	216,000
MidJourneyV6 (ter)	519,849	—	519,849
Co-Spy-Bench (Cheng et al., 2025)	550,000	—	550,000
Dalle3 (Egan et al., 2024)	1,193,805	—	1,193,805
Out-of-Distribution Existing Datasets with both Real and Synthetic Images			
FourierSpectrumDiscrepancies (Dzanic et al., 2020)	90	15	75
FakeInversion (Cazenavette et al., 2024)	1,300	650	650
UniversalFakeDetect (Ojha et al., 2023)	10,000	2,000	8,000
Dalle Recognition Dataset (Koliha, 2024)	21,635	3,780	17,855
Chameleon (Yan et al., 2024)	26,033	14,863	11,170
AI-GI-Detection-Quality-Paradox (Xiao et al., 2025)	27,864	3,864	24,000
DiTFake (Li et al., 2025b)	30,000	15,000	15,000
Diffusion1kSteps (Tan et al., 2024)	35,992	18,000	17,992
GANGen-Detection (Chuangchuang Tan, 2024)	36,000	18,000	18,000
RobustLDM (Rajan et al., 2024)	42,752	6,000	36,752
RealRobustBench (Li et al., 2025a)	53,999	26,999	27,000
LDMFakeDetect (Rajan and Lee, 2025)	61,352	6,000	55,352
DIF (Sinita and Fried, 2024)	75,344	37,672	37,672
ForenSynths (Wang et al., 2020)	90,329	45,169	45,160
DNF-TestSet (Zhang and Xu, 2025)	91,267	7,000	84,267
DeepFakeFace (Song et al., 2023)	120,000	30,000	90,000
AIGCDetectBench (Zhong et al., 2024)	152,597	76,298	76,299
AI-GI-Holmes (Zhou et al., 2025b)	164,996	83,850	81,146
AI-Artwork (Kholy, 2024)	271,993	81,444	190,549
DiffusionForensics (Wang et al., 2023)	313,368	86,000	227,368
B-Free (Guillaro et al., 2025)	361,584	52,482	309,102
LASTED (Wu et al., 2025a)	367,533	136,287	231,246
AIGIBench (Li et al., 2025c)	520,826	260,514	260,312
DeepFakeBench (Yan et al., 2023)	803,991	70,998	732,993
Percept-Lens Evaluation Suite	7,129,395	1,850,896	5,278,499

E. Extended Investigation of Generalization with Percept-Lens suite

Table 10 | **Closed-form Gaussian discriminants versus released AI-generated image detector heads under matched (prior, encoder) conditions.** Mean class accuracy CA (dataset-wise balanced accuracy where defined; Section 5.1), macro-averaged across Percept-Lens evaluation datasets. For each released detector checkpoint, we report (i) the released decision head (*Out-of-the-Shelf*) and (ii) closed-form baselines fitted on the *same frozen encoder features* using the corresponding public training prior: Euc-NCM (Wu et al., 2025b) and the Gaussian ladder. (Parentheses indicate the training subset reported by the original work when applicable.)

Detection Model	Out-of-the-Shelf	Euc-NCM (Wu et al., 2025b)	Cos-NCM	GNB	Mah-NCM	QDA
Trained with GenImage (Zhu et al., 2023) (Diffusion Model Images based on ImageNet-1k Dataset)						
DRCT-ConvNeXt (Full GenImage) (Chen et al., 2024)	46.92%	65.44%	64.67%	58.21%	65.54%	61.19%
Trained with DRCT-2M (Chen et al., 2024) (Stable Diffusion Model Images based on COCO Dataset)						
DRCT-ConvNeXt (DRCT-SDv1) (Chen et al., 2024)	46.92%	57.14%	57.64%	57.79%	52.66%	59.93%
DRCT-ConvNeXt (DRCT-SDv2) (Chen et al., 2024)	50.60%	62.02%	60.95%	61.82%	60.69%	61.20%
Trained with ELSA-D3 (Baraldi et al., 2025) (Diffusion Model Images based on LAION-400M Dataset)						
CoDE-SVM (ELSA-D3) (Baraldi et al., 2025)	36.01%	63.63%	62.70%	65.98%	66.15%	64.00%
CoDE-Linear (ELSA-D3) (Baraldi et al., 2025)	62.34%	63.63%	62.70%	65.98%	66.15%	64.00%

Table 11 | **Encoder scaling under a fixed prior: representation choice strongly affects transfer.** Mean class accuracy CA (macro-averaged across Percept-Lens evaluation datasets) for Gaussian discriminant heads conditioned on the same CommunityForensics training prior (Park and Owens, 2025), using frozen CLIP encoders of increasing capacity pretrained and available in OpenCLIP (Ilharco et al., 2021). Only the head is changed. No detector fine-tuning is performed.

Frozen Encoders	Euc-NCM (Wu et al., 2025b)	Cos-NCM	GNB	Mah-NCM	QDA
convnext_base-laion400m_s13b_b51k	65.64%	65.00%	65.99%	67.96%	62.27%
convnext_base_w-laion_aesthetic_s13b_b82k	67.26%	67.25%	66.97%	73.32%	64.98%
convnext_base_w-laion2b_s13b_b82k_augreg	67.64%	67.66%	68.20%	73.43%	66.93%
convnext_base_w-laion2b_s13b_b82k	66.95%	66.91%	66.00%	74.43%	63.32%
convnext_base_w_320-laion_aesthetic_s13b_b82k	67.22%	67.16%	66.88%	72.75%	64.89%
convnext_base_w_320-laion_aesthetic_s13b_b82k_augreg	67.33%	66.99%	66.43%	73.06%	64.52%
convnext_large_d-laion2b_s26b_b102k_augreg	68.99%	68.73%	68.58%	74.81%	67.48%
convnext_large_d_320-laion2b_s29b_b131k_ft	69.11%	68.30%	68.13%	74.87%	67.91%
convnext_large_d_320-laion2b_s29b_b131k_ft_soup	68.49%	67.77%	67.25%	73.71%	66.45%
convnext_xxlarge-laion2b_s34b_b82k_augreg	71.61%	70.73%	72.14%	78.76%	71.70%
convnext_xxlarge-laion2b_s34b_b82k_augreg_soup	71.50%	70.77%	71.89%	79.34%	71.44%
convnext_xxlarge-laion2b_s34b_b82k_augreg_rewind	71.60%	70.75%	71.94%	79.41%	71.34%
ViT-B-16-datacomp_xl_s13b_b90k	67.26%	67.57%	67.48%	72.27%	64.20%
ViT-B-32-datacomp_xl_s13b_b90k	65.05%	66.13%	66.19%	72.23%	62.56%
ViT-L-14-datacomp_xl_s13b_b90k	64.83%	64.73%	63.74%	66.39%	64.68%
ViT-L-14-laion400m_e32	65.93%	65.14%	65.64%	71.90%	63.39%
ViT-L-14-laion400m_e31	65.91%	65.19%	65.61%	71.92%	63.40%
ViT-L-14-laion2b_s32b_b82k	66.67%	66.33%	65.46%	73.06%	62.95%
ViT-L-14-commonpool_xl_clip_s13b_b90k	67.31%	69.42%	68.06%	74.60%	65.43%
ViT-L-14-commonpool_xl_laion_s13b_b90k	70.22%	70.81%	71.09%	75.13%	66.23%
ViT-L-14-quickgelu-metaclip_400m	66.73%	70.09%	71.15%	75.74%	67.72%
ViT-L-14-quickgelu-metaclip_fullcc	69.45%	72.62%	72.65%	76.07%	68.22%
ViT-L-14-quickgelu-dfn2b	65.06%	70.29%	68.69%	76.21%	66.36%
ViT-L-14-dfn2b_s39b	65.86%	70.70%	69.92%	76.24%	67.99%
ViT-L-14-commonpool_xl_s13b_b90k	67.34%	70.13%	72.37%	78.77%	69.05%
ViT-L-14-336-quickgelu-openai	68.20%	70.42%	72.33%	78.85%	71.36%
ViT-H-14-quickgelu-metaclip_fullcc	71.58%	74.11%	73.68%	77.83%	69.08%
ViT-H-14-quickgelu-dfn5b	70.80%	74.92%	74.00%	79.02%	72.82%
ViT-H-14-worldwide-quickgelu-metaclip2_worldwide	79.15%	81.57%	84.01%	84.20%	76.89%
ViT-H-14-378-quickgelu-dfn5b	72.12%	76.05%	74.75%	78.36%	74.02%
ViT-H-14-worldwide-378-metaclip2_worldwide	82.10%	84.81%	86.86%	85.53%	80.68%
ViT-bigG-14-quickgelu-metaclip_fullcc	72.62%	76.43%	76.65%	79.56%	72.17%
ViT-bigG-14-worldwide-metaclip2_worldwide	85.79%	85.72%	86.16%	87.36%	80.28%
ViT-bigG-14-worldwide-378-metaclip2_worldwide	88.65%	88.94%	89.01%	89.28%	83.77%

Continued on next page

Frozen Encoders	Euc-NCM (Wu et al., 2025b)	Cos-NCM	GNB	Mah-NCM	QDA
-----------------	----------------------------------	---------	-----	---------	-----

Table 12 | **Pretraining objective affects transfer even without detector training.** Mean class accuracy CA (macro-averaged across Percept-Lens evaluation datasets) for Gaussian discriminant heads conditioned on the same CommunityForensics training prior (Park and Owens, 2025), using frozen encoders trained with different objectives.

Frozen Encoders	Euc-NCM (Wu et al., 2025b)	Cos-NCM	GNB	Mah-NCM	QDA
CLIP-RoBERTa	63.81%	64.64%	63.87%	69.10%	62.00%
CLIP-XLM-RoBERTa-Base	63.13%	64.34%	64.47%	72.20%	62.34%
MAE-Base	58.35%	58.13%	59.37%	68.46%	63.74%
MAE-Large	59.56%	59.72%	62.39%	68.39%	60.79%
BEiT-Base	57.81%	57.69%	58.30%	59.57%	52.12%
SigLIP-so400m	56.30%	57.78%	58.55%	64.02%	59.01%
SigLIP2-Base-224	58.37%	58.37%	58.34%	61.78%	55.37%
SigLIP2-Base-512	58.58%	58.79%	58.68%	63.28%	56.59%
SigLIP2-Large-512	56.10%	56.03%	56.92%	69.30%	60.75%
coca_ViT-L-14]	66.90%	67.01%	64.00%	70.81%	63.43%
coca_ViT-L-14-mscoco_finetuned	62.06%	61.51%	62.81%	69.50%	64.80%
EVA01-g-14-laion400m_s11b_b41k	69.54%	68.51%	66.61%	76.74%	62.44%
EVA01-g-14-plus-merged2b_s11b_b114k	69.56%	71.17%	70.12%	78.72%	67.08%
EVA02-B-16-merged2b_s8b_b131k	63.75%	63.08%	64.53%	71.38%	60.78%
EVA02-L-14-merged2b_s4b_b131k	66.64%	67.16%	67.08%	72.81%	63.06%
CLIPA-ViT-L-14	65.69%	66.90%	66.31%	76.08%	68.51%
CLIPA-ViT-L-14-336	65.42%	66.92%	67.08%	76.25%	69.25%
CLIPA-ViT-H-14	67.23%	68.18%	69.12%	79.17%	70.58%
CLIPA-ViT-bigG-14-336	68.66%	70.67%	72.72%	80.27%	73.00%
nllb-clip-large-v1	70.73%	69.71%	70.61%	77.65%	65.80%
nllb-clip-large-siglip-v1	59.72%	59.48%	59.24%	67.55%	63.01%
nllb-clip-large-siglip-mrl	59.21%	59.02%	59.34%	68.23%	62.47%
MobileCLIP2-L-14-dfndr2b	66.48%	69.43%	68.33%	75.12%	65.68%
ViTamin-S	63.22%	64.43%	63.51%	67.72%	65.09%
ViTamin-S-LTT	64.41%	65.18%	64.86%	68.38%	65.07%
ViTamin-B	62.75%	64.25%	64.21%	70.75%	65.65%
ViTamin-B-LTT	64.97%	65.60%	64.85%	72.40%	67.23%
ViTamin-L	67.94%	68.42%	69.08%	76.64%	69.57%
ViTamin-L-256	67.57%	68.19%	68.07%	75.90%	70.56%
ViTamin-L-336	67.36%	68.36%	67.76%	76.18%	69.49%
ViTamin-L-384	74.43%	73.08%	76.17%	76.35%	65.83%
ViTamin-L2	70.11%	69.25%	70.75%	77.03%	70.16%
ViTamin-L2-256	70.50%	69.74%	70.01%	77.18%	71.64%
ViTamin-L2-336	70.42%	69.62%	69.72%	77.07%	69.77%
ViTamin-L2-384	70.53%	69.82%	68.51%	75.64%	66.00%
ViTamin-XL-256	71.25%	70.35%	71.62%	80.74%	71.36%
ViTamin-XL-336	71.63%	71.10%	71.54%	79.92%	70.52%
ViTamin-XL-384	71.56%	70.91%	72.49%	79.82%	71.52%
DINOv2-Small	62.02%	61.77%	60.99%	66.63%	60.17%
DINOv2-Base	63.79%	64.06%	62.08%	68.45%	59.43%

Continued on next page

Frozen Encoders	Euc-NCM (Wu et al., 2025b)	Cos-NCM	GNB	Mah-NCM	QDA
DINOv2-Large	63.03%	62.78%	62.29%	70.17%	59.34%
DINOv3-ConvNeXt-Tiny	61.14%	62.31%	59.97%	65.68%	59.85%
DINOv3-ConvNeXt-Small	61.70%	61.98%	59.60%	65.65%	60.30%
DINOv3-ConvNeXt-Base	64.54%	63.46%	61.39%	70.34%	62.68%
DINOv3-ConvNeXt-Large	69.32%	67.70%	69.02%	74.15%	66.77%
DINOv3-ViT-S-16	64.52%	64.26%	64.18%	66.92%	61.55%
DINOv3-ViT-S-16+	64.46%	64.03%	63.78%	67.35%	63.35%
DINOv3-ViT-B-16	68.83%	66.69%	69.67%	76.08%	65.32%
DINOv3-ViT-L-16	79.34%	74.74%	79.32%	83.31%	71.37%
DINOv3-ViT-H-16+	82.13%	79.58%	80.99%	86.22%	77.52%
PE-Core-T-16-384-meta	54.56%	54.78%	54.54%	63.27%	55.27%
PE-Core-S-16-384-meta	64.32%	63.01%	64.63%	68.13%	63.69%
PE-Core-B-16-meta	67.25%	67.02%	68.19%	73.08%	66.10%
PE-Core-L-14-336-meta	81.78%	80.77%	84.76%	89.51%	78.76%

Table 10 extends the matched (prior, encoder) audit to additional ConvNeXt checkpoints (Chen et al., 2024) and alternative CoDE heads (Baraldi et al., 2025). The pattern remains mostly unfavorable to the released heads. On the same frozen ConvNeXt representation, replacing the released head with a closed-form Gaussian rule yields CA gains for all three DRCT checkpoints: from 46.92% to 65.54% for the full GenImage model, from 46.92% to 59.93% for DRCT-SDv1, and from 50.60% to 62.02% for DRCT-SDv2. The CoDE comparison isolates the head effect more directly. CoDE-Linear and CoDE-SVM induce identical closed-form baselines because they share the same frozen encoder and the same support prior, yet their released heads differ sharply (62.34% versus 36.01%). Representation quality alone cannot explain this gap. The comparison instead points to head sensitivity on top of a fixed representation. As throughout the paper, this remains a post hoc swap on the final released representation, not a replay of the original training pipeline. The conservative conclusion is therefore that the released head does not always recover separability already present in the final encoder, not that end-to-end training is unnecessary.

Table 11 shows that the encoder effect is not confined to the smaller OpenAI sweep in the main text. Under the same CommunityForensics prior and with no detector fine-tuning, the best CA rises from 67.96% for convnext_base-laion400m_s13b_b51k to 79.41% for convnext_xlarge-laion2b_s34b_b82k_augreg_rewind. The larger ViT family exhibits a wider spread, from 66.39% to 89.28%. This spread is larger than many method-level gains reported in the detector literature, making head-only comparisons difficult to interpret unless encoder choice is controlled. Mah-NCM is the modal winner across rows, but the pattern is not universal. On the strongest encoders the margin often collapses, and in isolated cases another Gaussian rule is competitive or better. Any claim of universal Mahalanobis dominance would therefore be overstated.

Table 12 further reinforces the representation-centric reading. The best CA ranges from 59.57% for Beit-Base to 89.51% for PE-Core-L-14-336-meta, while DINOv3-ViT-H16plus already reaches 86.22% without any detector-specific fine-tuning. These gaps are large enough that they should not be treated as marginal implementation details. At the same time, this table should not be over-interpreted as a clean causal statement about a single pretraining objective, because objective, scale, data, and architecture all vary together. The supported claim is narrower. The choice of frozen encoder feature space is a first-order determinant of transfer, and a simple moment-based head on a strong encoder can outperform a specialized detector head trained on a weaker one. Overall, the appendix supports a

system-level conclusion rather than an algorithmic one. Cross-dataset AIGI detection depends strongly on the joint choice of support prior and representation, while the downstream head often contributes less than its standalone presentation suggests.

Table 13 | **QuickGELU versus GELU in otherwise matched encoders.** Mean class accuracy CA (macro-averaged across Percept-Lens evaluation datasets) for Gaussian discriminant heads conditioned on the same CommunityForensics training prior (Park and Owens, 2025), using frozen encoders that differ primarily in whether QuickGELU is used.

Frozen Encoders	Euc-NCM (Wu et al., 2025b)	Cos-NCM	GNB	Mah-NCM	QDA
ViT-B-16-openai	63.39%	64.66%	64.98%	73.19%	64.65%
ViT-B-16-quickgelu-openai	64.66%	64.65%	66.37%	74.18%	65.70%
ViT-B-32-openai	63.44%	63.95%	64.34%	69.07%	62.20%
ViT-B-32-quickgelu-openai	64.53%	64.25%	64.83%	70.61%	62.81%
ViT-L-14-openai	64.19%	66.96%	68.37%	76.87%	69.02%
ViT-L-14-quickgelu-openai	66.18%	68.44%	69.79%	76.96%	68.87%
ViT-L-14-336-openai	66.33%	69.05%	70.47%	78.89%	71.14%
ViT-L-14-336-quickgelu-openai	68.20%	70.42%	72.33%	78.85%	71.36%
ViT-L-14-metaclip_400m	64.85%	69.76%	70.40%	74.25%	66.65%
ViT-L-14-quickgelu-metaclip_400m	66.73%	70.09%	71.15%	75.74%	67.72%
ViT-L-14-metaclip_fullcc	68.07%	72.22%	72.81%	75.76%	67.51%
ViT-L-14-quickgelu-metaclip_fullcc	69.45%	72.62%	72.65%	76.07%	68.22%
ViT-L-14-dfn2b	66.91%	71.24%	70.16%	75.83%	66.52%
ViT-L-14-quickgelu-dfn2b	65.06%	70.29%	68.69%	76.21%	66.36%
ViT-H-14-worldwide-metaclip2_worldwide	78.59%	80.28%	82.89%	83.59%	76.55%
ViT-H-14-worldwide-quickgelu-metaclip2_worldwide	79.15%	81.57%	84.01%	84.20%	76.89%
ViT-H-14-378-dfn5b	73.28%	74.43%	74.08%	78.43%	72.76%
ViT-H-14-378-quickgelu-dfn5b	72.12%	76.05%	74.75%	78.36%	74.02%

Table 13 indicates that activation choice is a secondary factor relative to encoder family and scale. Across most matched pairs, the change in CA is modest, and the sign is not consistent across families. QuickGELU helps several models, but it does not provide a plausible explanation for the much larger performance differences observed across encoder families in Tables 11 and 12.

Table 14 shows that the aggregate CA gains are broadly distributed across evaluation datasets rather than being driven by a small subset of low-difficulty benchmarks. Mah-NCM is strongest on many mixed and synthetic-only benchmarks, but the exceptions are informative: QDA is best on CelebA-Spoof (Zhang et al., 2020), GNB is strongest on DiffusionForensics (Wang et al., 2023) and DeepFakeBench (Yan et al., 2023), and cosine or Euclidean rules slightly lead on a small number of synthetic-only datasets such as SPAI (Karageorgiou et al., 2025) and MidJourneyV6 (ter). The dataset-level view supports the same restrained conclusion as the averaged tables. Covariance-aware scoring is often useful, but no single Gaussian assumption is uniformly best across all shifts.

F. Threshold-Free (AUC) Evaluation on Mixed Datasets in Percept-Lens suite

In addition to CA, we report ROC-AUC on the 24 mixed datasets that contain both real and synthetic images (Table 9). AUC is not a universally stronger metric than CA, but it is a cleaner diagnostic of ranking quality because it is threshold-free and excludes one-class datasets. If the gains from the Gaussian ladder were mainly artifacts of a favorable operating point, they should shrink under AUC. In several cases they do not, although the counterexamples also become more informative.

Table 15 preserves the central matched-head finding under a calibration-agnostic metric. The size of the improvements suggests that threshold selection alone does not explain the result. CoDE-SVM rises from 0.3061 to 0.7039, DRCT-ConvNeXt on full GenImage from 0.4977 to 0.7417, AIDE on CNNSpot from 0.4917 to 0.6787, and AIDE on full GenImage from 0.5389 to 0.7082. These are ranking

Table 14 | **Per-dataset class accuracy under a fixed encoder and varying training prior.** Class accuracy CA for individual datasets in the Percept-Lens evaluation suite for Gaussian discriminant heads under the fixed frozen PE-Core-bigG-14-448 encoder (Bolya et al., 2025) and the same CommunityForensics support prior (Park and Owens, 2025) throughout.

Evaluation Dataset	Euc-NCM (Wu et al., 2025b)	Cos-NCM	GNB	Mah-NCM	QDA
Out-of-Distribution Existing Datasets with only Real Images					
Unsplash (Lite Subset) (uns)	24.13%	29.07%	75.56%	99.07%	96.20%
InstagramImagesWithCaptions (ins)	99.65%	99.40%	99.96%	99.96%	99.98%
Anime Faces Dataset (Churchill and Chao, 2019)	100.00%	100.00%	99.98%	98.20%	100.00%
Anime Images (Banana_Leopard, 2023)	99.95%	99.96%	99.97%	99.89%	100.00%
CelebA-Spoof (Zhang et al., 2020)	73.77%	68.03%	79.56%	84.70%	94.55%
Out-of-Distribution Existing Datasets with only Synthetic Images					
AGIQA-1k (Zhang et al., 2023)	99.81%	99.91%	98.80%	100.00%	97.41%
AGIQA-3k (Li et al., 2024)	85.81%	84.51%	75.52%	91.15%	51.84%
SPAI (Karageorgiou et al., 2025)	97.72%	97.77%	90.82%	92.00%	76.58%
SynthBuster Extended (Bammey, 2024; Guillaro et al., 2025)	99.95%	99.94%	98.52%	99.05%	91.43%
SynthScars (Kang et al., 2025)	79.61%	79.17%	62.32%	91.50%	50.90%
GigaGAN (Kang et al., 2023)	94.86%	94.33%	90.27%	94.31%	92.08%
LatentDiffusion (Corvi et al., 2023)	98.16%	97.85%	99.16%	99.86%	98.42%
MidJourneyV6 (ter)	99.51%	99.49%	93.92%	96.99%	83.88%
Co-Spy-Bench (Cheng et al., 2025)	99.66%	99.71%	98.27%	99.29%	95.88%
Dalle3 (Egan et al., 2024)	92.22%	93.28%	42.49%	99.25%	59.02%
Out-of-Distribution Existing Datasets with both Real and Synthetic Images					
FourierSpectrumDiscrepancies (Dzanic et al., 2020)	85.33%	86.00%	88.67%	100.00%	97.33%
FakeInversion (Cazenavette et al., 2024)	85.08%	84.38%	91.38%	99.00%	96.08%
UniversalFakeDetect (Ojha et al., 2023)	98.06%	98.15%	99.07%	99.56%	97.55%
Dalle Recognition Dataset (Koliha, 2024)	88.56%	88.55%	86.59%	97.16%	83.96%
Chameleon (Yan et al., 2024)	79.51%	79.20%	72.03%	90.52%	59.45%
AIGI-Detection-Quality-Paradox (Xiao et al., 2025)	83.70%	82.56%	92.09%	99.42%	92.71%
DiTFake (Li et al., 2025b)	92.61%	91.91%	98.28%	99.53%	97.93%
Diffusion1kSteps (Tan et al., 2024)	81.83%	81.17%	88.27%	93.80%	89.57%
GANGen-Detection (Chuangchuang Tan, 2024)	81.13%	82.48%	94.21%	98.10%	93.40%
RobustLDM (Rajan et al., 2024)	89.28%	87.62%	88.01%	98.22%	85.73%
RealRobustBench (Li et al., 2025a)	77.61%	76.96%	75.28%	95.26%	71.61%
LDMFakeDetect (Rajan and Lee, 2025)	88.85%	87.18%	88.39%	97.63%	84.54%
DIF (Sinita and Fried, 2024)	89.61%	88.51%	96.86%	98.62%	97.80%
ForenSynths (Wang et al., 2020)	75.85%	74.99%	84.11%	97.06%	91.78%
DNF-TestSet (Zhang and Xu, 2025)	98.67%	98.50%	99.43%	99.73%	98.92%
DeepFakeFace (Song et al., 2023)	59.90%	61.67%	62.08%	76.50%	57.65%
AIGCDetectBench (Zhong et al., 2024)	92.66%	92.22%	97.02%	98.57%	96.47%
AIGI-Holmes (Zhou et al., 2025b)	90.72%	89.28%	95.21%	98.97%	95.92%
AI-Artwork (Kholy, 2024)	88.42%	86.86%	90.20%	95.72%	84.95%
DiffusionForensics (Wang et al., 2023)	95.10%	95.19%	96.44%	91.16%	93.02%
B-Free (Guillaro et al., 2025)	87.75%	87.79%	85.35%	85.78%	85.13%
LASTED (Wu et al., 2025a)	75.98%	74.74%	80.46%	80.80%	76.27%
AIGIBench (Li et al., 2025c)	86.55%	85.88%	93.92%	95.21%	92.61%
DeepFakeBench (Yan et al., 2023)	51.39%	52.10%	54.59%	52.40%	51.08%

improvements, not merely operating-point adjustments. The table also contains counterexamples, since the released CF-224 and CF-384 heads remain marginally stronger than any closed-form surrogate on AUC. CA and AUC therefore diagnose different failure modes. A head swap can improve thresholded balanced accuracy while leaving global ranking unchanged, or vice versa. The strongest result in this block is obtained without detector fine-tuning: PE-Core-bigG-14-448 with Mah-NCM reaches 0.9693 AUC. This is not an apples-to-apples replacement for a released detector with a different backbone, but it shows how quickly the comparison shifts once the encoder feature space itself becomes stronger.

Table 15 | **Closed-form Gaussian discriminants versus released AI-generated image detector heads under matched (prior, encoder) conditions.** Mean ROC-Area Under Curve AUC (dataset-wise ROC-AUC where defined; Section 5.1), macro-averaged across the 24 mixed datasets in Percept-Lens evaluation datasets. For each released detector checkpoint, we report (i) the released decision head (*Out-of-the-Shelf*) and (ii) closed-form baselines fitted on the *same frozen encoder features* using the corresponding public training prior: Euc-NCM (Wu et al., 2025b) and the Gaussian ladder. (Parentheses indicate the training subset reported by the original work when applicable.)

Detection Model	Out-of-the-Shelf	Euc-NCM (Wu et al., 2025b)	Cos-NCM	GNB	Mah-NCM	QDA
Trained with CNNSpot (Wang et al., 2020) (ProGAN Images based on LSUN Dataset)						
UnivFD (Ojha et al., 2023)	0.7091	0.6995	0.6756	0.7145	0.7454	0.6994
AIDE (Yan et al., 2024)	0.4917	0.5256	0.5136	0.5278	0.6787	0.6383
Trained with GenImage (Zhu et al., 2023) (Diffusion Model Images based on ImageNet-1k Dataset)						
AIDE (GenImage-SDv1) (Yan et al., 2024)	0.5231	0.5620	0.5866	0.5710	0.6875	0.5929
Effort (GenImage-SDv1) (Yan et al., 2025)	0.7729	0.7951	0.8348	0.8339	0.8635	0.8196
DRCT-ConvNeXt (Full GenImage) (Chen et al., 2024)	0.4977	0.7270	0.7304	0.6176	0.7417	0.6936
DRCT-UnivFD (Full GenImage) (Chen et al., 2024)	0.7368	0.7488	0.8083	0.7801	0.8395	0.7789
AIDE (Full GenImage) (Yan et al., 2024)	0.5389	0.5240	0.5662	0.5322	0.7082	0.5850
Trained with DRCT-2M (Chen et al., 2024) (Stable Diffusion Model Images based on COCO Dataset)						
DRCT-ConvNeXt (DRCT-SDv1) (Chen et al., 2024)	0.4977	0.6194	0.6274	0.6236	0.5920	0.6460
DRCT-UnivFD (DRCT-SDv1) (Chen et al., 2024)	0.7359	0.6792	0.7055	0.7060	0.7389	0.7363
DRCT-ConvNeXt (DRCT-SDv2) (Chen et al., 2024)	0.5230	0.6569	0.6449	0.6704	0.6786	0.6542
DRCT-UnivFD (DRCT-SDv2) (Chen et al., 2024)	0.7366	0.6781	0.6916	0.7022	0.7527	0.7244
Trained with ELSA-D3 (Baraldi et al., 2025) or CommunityForensics (Park and Owens, 2025) (Diffusion Model Images based on LATON-400M Dataset)						
CoDE-SVM (ELSA-D3) (Baraldi et al., 2025)	0.3061	0.6986	0.7039	0.7034	0.6866	0.6714
CoDE-Linear (ELSA-D3) (Baraldi et al., 2025)	0.7065	0.6986	0.7039	0.7034	0.6866	0.6714
CoDE-kNN (ELSA-D3) (Baraldi et al., 2025)	0.6360	0.6986	0.7039	0.7034	0.6866	0.6714
CF-224 (CommunityForensics) (Park and Owens, 2025)	0.9131	0.8799	0.9101	0.9007	0.9093	0.8826
CF-384 (CommunityForensics) (Park and Owens, 2025)	0.9351	0.8821	0.9304	0.9327	0.9283	0.9320
Frozen image encoder (no detection fine-tuning)						
PE-Core-bigG-14-448 (Bolya et al., 2025)	—	0.8948	0.9309	0.9412	0.9693	0.9295

Table 16 | **Encoder scaling under a fixed prior: representation choice strongly affects transfer.** Mean ROC-Area Under Curve AUC (macro-averaged across the 24 mixed datasets in Percept-Lens evaluation datasets) for Gaussian discriminant heads conditioned on the same CommunityForensics training prior (Park and Owens, 2025), using frozen CLIP encoders of increasing capacity pretrained by OpenAI. Only the head is changed. No detector fine-tuning is performed.

Frozen Encoders	Euc-NCM (Wu et al., 2025b)	Cos-NCM	GNB	Mah-NCM	QDA
ResNet-50 (Timm, 2024a)	0.6066	0.5682	0.6370	0.7406	0.5844
ResNet-101 (He et al., 2016)	0.5999	0.5823	0.6157	0.7194	0.5990
ResNet-50x4 (Timm, 2024c)	0.6268	0.6150	0.6491	0.7633	0.6031
ResNet-50x16 (Timm, 2024b)	0.6519	0.6358	0.6829	0.8143	0.6404
ResNet-50x64 (Timm, 2024d)	0.6870	0.6752	0.6798	0.8345	0.6728
ViT-B/16 (Laion, 2024d)	0.6334	0.5965	0.6551	0.7823	0.6573
ViT-B/32 (Laion, 2024e)	0.6065	0.5831	0.6131	0.7147	0.6098
ViT-L/14 (Radford et al., 2021)	0.6654	0.6911	0.7165	0.8193	0.7051

Table 17 | **Encoder scaling under a fixed prior: representation choice strongly affects transfer.** Mean ROC-Area Under Curve AUC (macro-averaged across the 24 mixed datasets in Percept-Lens evaluation datasets) for Gaussian discriminant heads conditioned on the same CommunityForensics training prior (Park and Owens, 2025), using frozen CLIP encoders of increasing capacity pretrained and available in OpenCLIP (Ilharco et al., 2021). Only the head is changed. No detector fine-tuning is performed.

Frozen Encoders	Euc-NCM (Wu et al., 2025b)	Cos-NCM	GNB	Mah-NCM	QDA
convnext_base-laion400m_s13b_b51k	0.6288	0.6177	0.6486	0.7076	0.6254
convnext_base_w-laion_aesthetic_s13b_b82k	0.6820	0.6733	0.7028	0.7791	0.6491
convnext_base_w-laion2b_s13b_b82k_augreg	0.6775	0.6765	0.6940	0.8036	0.6298
convnext_base_w-laion2b_s13b_b82k	0.6528	0.6479	0.6814	0.7889	0.6262
convnext_base_w_320-laion_aesthetic_s13b_b82k	0.6752	0.6647	0.6833	0.7833	0.6509
convnext_base_w_320-laion_aesthetic_s13b_b82k_augreg	0.6694	0.6623	0.6629	0.7958	0.6505
convnext_large_d-laion2b_s26b_b102k_augreg	0.7101	0.7092	0.7038	0.8098	0.6545
convnext_large_d_320-laion2b_s29b_b131k_ft	0.6930	0.6866	0.6891	0.8059	0.6667
convnext_large_d_320-laion2b_s29b_b131k_ft_soup	0.7106	0.7038	0.7006	0.8155	0.6695
convnext_xxlarge-laion2b_s34b_b82k_augreg	0.7460	0.7446	0.7772	0.8653	0.7056
convnext_xxlarge-laion2b_s34b_b82k_augreg_soup	0.7427	0.7435	0.7722	0.8636	0.7033
convnext_xxlarge-laion2b_s34b_b82k_augreg_rewind	0.7452	0.7459	0.7755	0.8635	0.7036
ViT-B-16-datacomp_xl_s13b_b90k	0.6912	0.6811	0.7171	0.7993	0.6481
ViT-B-32-datacomp_xl_s13b_b90k	0.6258	0.6385	0.6563	0.7535	0.5838
ViT-L-14-datacomp_xl_s13b_b90k	0.7217	0.7036	0.7254	0.7577	0.7326
ViT-L-14-laion400m_e32	0.6461	0.6460	0.6524	0.7590	0.6300
ViT-L-14-laion400m_e31	0.6452	0.6448	0.6525	0.7598	0.6297
ViT-L-14-laion2b_s32b_b82k	0.6481	0.6535	0.6588	0.8017	0.6132
ViT-L-14-commonpool_xl_clip_s13b_b90k	0.7146	0.7361	0.7329	0.8228	0.6497
ViT-L-14-commonpool_xl_laion_s13b_b90k	0.7408	0.7493	0.7564	0.8109	0.6491
ViT-L-14-quickgelu-metaclip_400m	0.7089	0.7473	0.7639	0.8186	0.6867
ViT-L-14-quickgelu-metaclip_fullcc	0.7589	0.8029	0.8014	0.8394	0.6774
ViT-L-14-quickgelu-dfn2b	0.7033	0.7388	0.7357	0.8239	0.6526
ViT-L-14-dfn2b_s39b	0.6766	0.7611	0.7632	0.8353	0.6680
ViT-L-14-commonpool_xl_s13b_b90k	0.7218	0.7670	0.7810	0.8336	0.6842
ViT-L-14-336-quickgelu-openai	0.7046	0.7230	0.7598	0.8428	0.7269
ViT-H-14-quickgelu-metaclip_fullcc	0.8172	0.8249	0.8277	0.8559	0.6668
ViT-H-14-quickgelu-dfn5b	0.7820	0.8166	0.8145	0.8557	0.7048
ViT-H-14-worldwide-quickgelu-metaclip2_worldwide	0.8599	0.9142	0.9082	0.9211	0.7557
ViT-H-14-378-quickgelu-dfn5b	0.8095	0.8290	0.8282	0.8629	0.7339
ViT-H-14-worldwide-378-metaclip2_worldwide	0.8892	0.9334	0.9299	0.9274	0.8053
ViT-bigG-14-quickgelu-metaclip_fullcc	0.8107	0.8576	0.8469	0.8792	0.6965
ViT-bigG-14-worldwide-metaclip2_worldwide	0.8921	0.9320	0.9164	0.9328	0.7968
ViT-bigG-14-worldwide-378-metaclip2_worldwide	0.9169	0.9460	0.9353	0.9464	0.8399

Table 18 | **Pretraining objective affects transfer even without detector training.** Mean ROC-Area Under Curve AUC (macro-averaged across the 24 mixed datasets in Percept-Lens evaluation datasets) for Gaussian discriminant heads conditioned on the same CommunityForensics training prior (Park and Owens, 2025), using frozen encoders trained with different objectives.

Frozen Encoders	Euc-NCM (Wu et al., 2025b)	Cos-NCM	GNB	Mah-NCM	QDA
CLIP-RoBERTa	0.5812	0.5838	0.6208	0.7274	0.6025
CLIP-XLM-RoBERTa-Base	0.5985	0.6087	0.6391	0.7586	0.5921
CLIP-XLM-RoBERTa-Large (Cherti et al., 2023)	0.7264	0.7236	0.7261	0.8452	0.6492
MAE-Base	0.5251	0.5227	0.5334	0.6873	0.5770
MAE-Large	0.5434	0.5465	0.5769	0.7068	0.5672
MAE-Huge (He et al., 2022)	0.5401	0.5402	0.5806	0.7418	0.5738
BEiT-Base	0.5152	0.5060	0.5170	0.5715	0.5245
BEiT-Large (Bao et al., 2022)	0.5203	0.5128	0.5208	0.6062	0.5363
SigLIP-Large (Zhai et al., 2023)	0.5203	0.5294	0.5257	0.6474	0.5772
SigLIP-so400m	0.5228	0.5213	0.5248	0.6345	0.5530
SigLIP2-Base-224	0.5260	0.5337	0.5207	0.5998	0.5232
SigLIP2-Base-512	0.5209	0.5482	0.5175	0.6123	0.5238
SigLIP2-Large-512	0.5401	0.5684	0.5318	0.7056	0.5723
BLIP-Large (Li et al., 2022)	0.5473	0.5511	0.5378	0.7535	0.5591
BLIP2 (Li et al., 2023)	0.7187	0.7119	0.6359	0.8386	0.5871
coca_ViT-L-14	0.6654	0.6630	0.6859	0.8045	0.6089
coca_ViT-L-14-mscoco_finetuned	0.5681	0.5554	0.5892	0.7758	0.5818
EVA01-g-14-laion400m_s11b_b41k	0.7275	0.7218	0.6932	0.8275	0.5941
EVA01-g-14-plus-merged2b_s11b_b114k	0.7063	0.7609	0.7052	0.8461	0.6488
EVA02-B-16-merged2b_s8b_b131k	0.6305	0.6293	0.6337	0.7642	0.5929
EVA02-L-14-merged2b_s4b_b131k	0.7002	0.7281	0.6887	0.7993	0.6163
CLIPA-ViT-L-14	0.6955	0.7049	0.7193	0.8304	0.6739
CLIPA-ViT-L-14-336	0.7116	0.7223	0.7363	0.8350	0.6863
CLIPA-ViT-H-14	0.7073	0.7362	0.7500	0.8655	0.6997
CLIPA-ViT-bigG-14-336	0.7206	0.7871	0.7849	0.8828	0.7302
nllb-clip-large-v1	0.7265	0.7237	0.7260	0.8452	0.6488
nllb-clip-large-siglip-v1	0.5187	0.5094	0.5109	0.6906	0.5632
nllb-clip-large-siglip-mrl	0.4891	0.4827	0.5037	0.6968	0.5609
MobileCLIP2-L-14	0.7155	0.7313	0.7674	0.8415	0.6452
ViTamin-S	0.6070	0.6117	0.6403	0.7238	0.6345
ViTamin-S-LTT	0.6246	0.6379	0.6270	0.7425	0.6336
ViTamin-B	0.6547	0.6601	0.6679	0.7713	0.6643
ViTamin-B-LTT	0.6494	0.6542	0.6523	0.7848	0.6548
ViTamin-L	0.6823	0.7058	0.7334	0.8358	0.7092
ViTamin-L-256	0.6967	0.7180	0.7376	0.8435	0.7298
ViTamin-L-336	0.7084	0.7266	0.7482	0.8473	0.7259
ViTamin-L-384	0.7404	0.7594	0.7910	0.8814	0.6827
ViTamin-L2	0.7163	0.7207	0.7477	0.8500	0.7000
ViTamin-L2-256	0.7418	0.7416	0.7548	0.8596	0.7268
ViTamin-L2-336	0.7483	0.7439	0.7643	0.8671	0.7057
ViTamin-L2-384	0.7244	0.7270	0.7359	0.8297	0.6634
ViTamin-XL-256	0.7660	0.7589	0.7848	0.8770	0.7231

Continued on next page

Frozen Encoders	Euc-NCM (Wu et al., 2025b)	Cos-NCM	GNB	Mah-NCM	QDA
ViTamin-XL-336	0.7774	0.7724	0.7936	0.8759	0.7089
ViTamin-XL-384	0.7703	0.7643	0.7899	0.8816	0.7386
DINOv2-Small	0.6455	0.6561	0.6419	0.6827	0.5546
DINOv2-Base	0.6748	0.6835	0.6698	0.7275	0.5709
DINOv2-Large	0.6929	0.6928	0.6895	0.7576	0.5813
DINOv2-Giant (Oquab et al., 2024)	0.7007	0.7103	0.7021	0.8087	0.5896
DINOv3-ConvNeXt-Tiny	0.5989	0.6700	0.5966	0.6872	0.5748
DINOv3-ConvNeXt-Small	0.6480	0.6784	0.6103	0.7055	0.5865
DINOv3-ConvNeXt-Base	0.6884	0.6932	0.6506	0.7406	0.6179
DINOv3-ConvNeXt-Large	0.7519	0.7281	0.7554	0.8051	0.6768
DINOv3-ViT-S-16	0.6598	0.6546	0.6581	0.6788	0.5676
DINOv3-ViT-S-16+	0.6917	0.6827	0.6947	0.6973	0.6048
DINOv3-ViT-B-16	0.7608	0.7403	0.7697	0.8105	0.6485
DINOv3-ViT-L-16	0.8539	0.8064	0.8593	0.8869	0.7490
DINOv3-ViT-H-16+	0.8924	0.8447	0.8772	0.9111	0.8251
DINOv3-ViT-7B (Siméoni et al., 2025)	0.9177	0.8836	0.8931	0.9342	0.8078
PE-Core-T-16-384-meta	0.4960	0.4998	0.5057	0.6300	0.5215
PE-Core-S-16-384-meta	0.6095	0.5850	0.6270	0.7314	0.6162
PE-Core-B-16-meta	0.6527	0.6437	0.6879	0.8056	0.6356
PE-Core-L-14-336-meta	0.8758	0.8806	0.9039	0.9425	0.8130

Tables 16 and 17 confirms that the encoder effect survives under a threshold-free metric. Within the OpenAI CLIP family, the best AUC improves from 0.7406 for ResNet-50 to 0.8345 for ResNet-50x64, and from 0.7147 for ViT-B/32 to 0.8193 for ViT-L/14. The broader OpenCLIP sweep spans a wider range, from 0.7076 to 0.9464, with several large H/bigG models above 0.92. As in CA, Mah-NCM is the modal winner, but the advantage is not universal. On some of the strongest encoders the gap to cosine or diagonal rules becomes negligible, and after correcting row-wise maxima there are isolated reversals. These reversals argue against presenting any one covariance assumption as universally correct. The head family still matters, but less than representation choice in this sweep.

Table 18 shows a similarly large spread across encoder families, from 0.5715 for BeiT-Base to 0.9425 for PE-Core-L-14-336-meta, with DINOv3-ViT-7b already reaching 0.9342. Several masked-image or SigLIP-style models remain well below the strongest contrastive or distilled models. This should still be written cautiously. The table does not isolate objective alone, because data scale, architecture, and resolution also change. What it does establish is that pretraining family is a first-order determinant of transfer even before any detector-specific optimization is introduced. In particular, language alignment is not a sufficient explanation: some language-aligned encoders are middling, whereas some visually distilled representations are exceptionally strong.

Table 20 | **Training prior dominates transfer even under a fixed encoder.** Mean ROC-Area Under Curve AUC (macro-averaged across the 24 mixed datasets in Percept-Lens evaluation datasets) for Gaussian discriminant heads when varying only the *support prior* used to estimate moments, under the fixed frozen PE-Core-bigG-14-448 encoder (Bolya et al., 2025).

Training Dataset	Euc-NCM (Wu et al., 2025b)	Cos-NCM	GNB	Mah-NCM	QDA
CNNSpot (Wang et al., 2020)	0.8987	0.9196	0.9210	0.9290	0.6466
GenImage (Zhu et al., 2023)	0.8781	0.9624	0.9455	0.9589	0.9341
GenImage-SDv1	0.8974	0.9491	0.9467	0.9577	0.8407
DRCT-2M (Chen et al., 2024)	0.8505	0.9260	0.9177	0.8716	0.9300
DRCT-SDv1 (Chen et al., 2024)	0.8466	0.9508	0.9359	0.9372	0.9459
DRCT-SDv2 (Chen et al., 2024)	0.8504	0.9551	0.9427	0.9551	0.9471
ELSA-D3 (Baraldi et al., 2025)	0.9120	0.9363	0.9413	0.9645	0.9469
CommunityForensics (Park and Owens, 2025)	0.8948	0.9309	0.9412	0.9693	0.9295

Table 19 | **QuickGELU versus GELU under a threshold-free metric.** Mean ROC-Area Under Curve AUC (macro-averaged across the 24 mixed datasets in Percept-Lens evaluation datasets) for Gaussian discriminant heads conditioned on the same CommunityForensics training prior (Park and Owens, 2025), using frozen encoders that differ primarily in whether QuickGELU is used.

Frozen Encoder	Euc-NCM (Wu et al., 2025b)	Cos-NCM	GNB	Mah-NCM	QDA
ViT-B-16-openai	0.6310	0.6077	0.6498	0.7750	0.6500
ViT-B-16-quickgelu-openai	0.6334	0.5965	0.6551	0.7823	0.6573
ViT-B-32-openai	0.5819	0.5657	0.5928	0.6987	0.6027
ViT-B-32-quickgelu-openai	0.6065	0.5831	0.6131	0.7147	0.6098
ViT-L-14-openai	0.6708	0.6853	0.7134	0.8223	0.7012
ViT-L-14-quickgelu-openai	0.6654	0.6911	0.7165	0.8193	0.7051
ViT-L-14-metaclip_400m	0.6948	0.7266	0.7473	0.8088	0.6742
ViT-L-14-quickgelu-metaclip_400m	0.7089	0.7473	0.7639	0.8186	0.6867
ViT-L-14-metaclip_fullcc	0.7330	0.7898	0.7895	0.8347	0.6637
ViT-L-14-quickgelu-metaclip_fullcc	0.7589	0.8029	0.8014	0.8394	0.6774
ViT-L-14-dfn2b	0.7195	0.7480	0.7443	0.8188	0.6546
ViT-L-14-quickgelu-dfn2b	0.7033	0.7388	0.7357	0.8239	0.6526
ViT-L-14-336-openai	0.7115	0.7220	0.7537	0.8527	0.7253
ViT-L-14-336-quickgelu-openai	0.7046	0.7230	0.7598	0.8428	0.7269
ViT-H-14-worldwide-metaclip2	0.8506	0.9081	0.9027	0.9108	0.7573
ViT-H-14-worldwide-quickgelu-metaclip2	0.8599	0.9142	0.9082	0.9211	0.7557
ViT-H-14-378-dfn5b	0.7930	0.7988	0.8045	0.8558	0.7125
ViT-H-14-378-quickgelu-dfn5b	0.8095	0.8290	0.8282	0.8629	0.7339

Table 19 suggests that the choice between GELU and QuickGELU is secondary relative to encoder family and scale. Most matched pairs differ by less than two AUC points, and the sign of the difference is not consistent across families. We therefore treat activation choice as a modest architecture-level modifier rather than a primary explanation for transfer.

Table 20 shows that prior choice remains important even after fixing a strong encoder. The best AUC varies from 0.9290 with CNNSpot support to 0.9693 with CommunityForensics support. More importantly, the preferred Gaussian rule depends on the prior: CommunityForensics and ELSA-D3 favor Mah-NCM, DRCT-2M favors QDA or cosine, and GenImage favors cosine once the row-wise maximum is corrected. Thus, the covariance structure that transfers best is prior-conditioned rather than universal.

At the per-dataset level, the AUC table (Table 21) shows both the breadth and the limits of the

Table 21 | **Per-dataset AUC under a fixed encoder and varying training prior.** ROC-Area Under Curve AUC for individual datasets in Percept-Lens evaluation mixed datasets (24 datasets total) for Gaussian discriminant heads, under the fixed frozen PE-Core-bigG-14-448 encoder (Bolya et al., 2025) and the fixed CommunityForensics support prior (Park and Owens, 2025).

Training Dataset	Euc-NCM (Wu et al., 2025b)	Cos-NCM	GNB	Mah-NCM	QDA
FourierSpectrumDiscrepancies (Dzanic et al., 2020)	0.8569	0.8996	0.9551	1.0000	0.9956
FakeInversion (Cazenavette et al., 2024)	0.8714	0.9811	0.9588	0.9997	0.9844
UniversalFakeDetect (Ojha et al., 2023)	0.9900	0.9982	0.9980	0.9998	0.9906
Dalle Recognition Dataset (Koliha, 2024)	0.9087	0.9457	0.9528	0.9916	0.9708
Chameleon (Yan et al., 2024)	0.9548	0.9549	0.9674	0.9932	0.8537
AIGI-Detection-Quality-Paradox (Xiao et al., 2025)	0.8714	0.9668	0.9768	0.9992	0.9909
DiTFake (Li et al., 2025b)	0.9504	0.9989	0.9979	0.9995	0.9935
Diffusion1kSteps (Tan et al., 2024)	0.9025	0.9127	0.9586	0.9873	0.9234
GANGen-Detection (Chuangchuang Tan, 2024)	0.9891	0.9890	0.9939	0.9979	0.9876
RobustLDM (Rajan et al., 2024)	0.9378	0.9549	0.9659	0.9973	0.9397
RealRobustBench (Li et al., 2025a)	0.8445	0.8670	0.8820	0.9923	0.8071
LDMFakeDetect (Rajan and Lee, 2025)	0.9342	0.9509	0.9641	0.9959	0.9335
DIF (Sinita and Fried, 2024)	0.9184	0.9884	0.9918	0.9995	0.9904
ForenSynths (Wang et al., 2020)	0.7858	0.9413	0.9383	0.9947	0.9748
DNF-TestSet (Zhang and Xu, 2025)	0.9949	0.9978	0.9990	0.9999	0.9949
DeepFakeFace (Song et al., 2023)	0.8500	0.8490	0.8424	0.9346	0.8261
AIGCDetectBench (Zhong et al., 2024)	0.9463	0.9884	0.9897	0.9975	0.9806
AIGI-Holmes (Zhou et al., 2025b)	0.9348	0.9744	0.9892	0.9988	0.9860
AI-Artwork (Kholy, 2024)	0.9163	0.9383	0.9670	0.9913	0.9714
DiffusionForensics (Wang et al., 2023)	0.9701	0.9916	0.9905	0.9912	0.9505
B-Free (Guillaro et al., 2025)	0.9140	0.9373	0.9362	0.9439	0.9294
LASTED (Wu et al., 2025a)	0.7766	0.8022	0.8250	0.8008	0.7916
AIGIBench (Li et al., 2025c)	0.8941	0.9539	0.9817	0.9843	0.9545
DeepFakeBench (Yan et al., 2023)	0.5620	0.5595	0.5666	0.6735	0.5865

Table 22 | **Data-efficiency of moment-based heads.** Mean ROC-Area Under Curve AUC (macro-averaged across the 24 mixed datasets in Percept-Lens evaluation datasets) for Gaussian discriminant heads as the support set used to estimate moments is subsampled from CommunityForensics (Park and Owens, 2025), under the fixed frozen PE-Core-bigG-14-448 encoder (Bolya et al., 2025). (Superscript/subscript denote max/min deviations (in percentage points) from the reported mean).

Amount	Euc-NCM (Wu et al., 2025b)	Cos-NCM	GNB	Mah-NCM	QDA
0.001%	0.8345 ^{+0.0224} _{-0.0407}	0.9356 ^{+0.0037} _{-0.0042}	0.9239 ^{+0.0134} _{-0.0185}	0.8788 ^{+0.0132} _{-0.0187}	0.9025 ^{+0.0277} _{-0.0148}
0.005%	0.8852 ^{+0.0052} _{-0.0032}	0.9313 ^{+0.0041} _{-0.0036}	0.9379 ^{+0.0039} _{-0.0032}	0.9383 ^{+0.0037} _{-0.0049}	0.9089 ^{+0.0122} _{-0.0137}
0.01%	0.8897 ^{+0.0080} _{-0.0044}	0.9319 ^{+0.0018} _{-0.0029}	0.9396 ^{+0.0052} _{-0.0048}	0.9469 ^{+0.0061} _{-0.0045}	0.8991 ^{+0.0166} _{-0.0197}
0.1%	0.8919 ^{+0.0050} _{-0.0036}	0.9303 ^{+0.0036} _{-0.0032}	0.9403 ^{+0.0015} _{-0.0013}	0.9638 ^{+0.0017} _{-0.0019}	0.8308 ^{+0.0044} _{-0.0119}
1%	0.8947 ^{+0.0010} _{-0.0012}	0.9310 ^{+0.0009} _{-0.0010}	0.9410 ^{+0.0002} _{-0.0001}	0.9668 ^{+0.0014} _{-0.0008}	0.8478 ^{+0.0035} _{-0.0054}
100%	0.8948	0.9309	0.9412	0.9693	0.9295

average improvement. Mah-NCM reaches near-ceiling AUC on many mixed benchmarks, including RealRobustBench (Li et al., 2025a) (0.9923), ForenSynths (Wang et al., 2020) (0.9947), and AIGI-Holmes (Zhou et al., 2025b) (0.9988), so the macro-average is not driven by a small subset of low-difficulty benchmarks. The failure cases are also informative. LASTED favors GNB, DiffusionForensics (Wang et al., 2023) is effectively tied between cosine and Mahalanobis, and DeepFakeBench (Yan et al., 2023) remains difficult for every head, peaking at only 0.6735. Comparing Table 21 with Table 14 is especially informative on DiffusionForensics: Mah-NCM is not best in CA but is essentially tied for best in AUC, suggesting that the representation contains ranking signal even though the default operating point is suboptimal.

Table 22 adds a small-data qualifier. At the smallest support size, simpler rules can be more reliable than shared full-covariance estimation. With only 0.001% of CommunityForensics, Cos-NCM already reaches 0.9356 AUC, whereas Mah-NCM drops to 0.8788. Mah-NCM overtakes once modest support is available, reaching 0.9638 at 0.1% and 0.9668 at 1%, close to the full-support 0.9693. QDA is visibly unstable and non-monotone under subsampling, consistent with a higher-variance class-specific covariance estimate. For few-shot adaptation, the data do not support the general rule “always use the richest Gaussian model.” They support the more specific rule “match the covariance model to the amount of support available.”

Overall, the AUC results do not merely replicate the CA results. The threshold-free view strengthens the main claim. Low-order feature geometry remains highly competitive, while the failure cases show that the residual value of a trained head, when it exists, is real but narrower than many standalone detector comparisons suggest.

G. Class-Conditional Gaussianity Diagnostics of Evaluation Features

A natural objection to the Gaussian ladder is that frozen-encoder features on shifted evaluation data need not be exactly Gaussian. The relevant object, however, is the *class-conditional* feature distribution $p_\phi(z \mid y = c, \mathcal{D})$, not the pooled distribution $p_\phi(z \mid \mathcal{D})$: even if each class-conditional component were Gaussian, their mixture would in general not be Gaussian. We therefore report class-conditional Gaussianity diagnostics for each dataset–class subset in feature space.

These quantities are *diagnostics*, not formal hypothesis tests. They summarize departures from Gaussian structure, but they do not provide calibrated p -values and should not be interpreted as accepting or rejecting exact normality.

Let $z_i = \phi(x_i) \in \mathbb{R}^d$ denote the frozen-encoder feature of sample x_i , restricted to a fixed dataset $\mathcal{D}$ and class $c \in \{0, 1\}$. With n samples in that subset, empirical mean $\bar{z}$, and regularized sample covariance

$$\hat{\Sigma} = \frac{1}{n-1} \sum_{i=1}^n (z_i - \bar{z})(z_i - \bar{z})^\top + \varepsilon I_d, \quad \varepsilon = 10^{-8}, \quad (\text{G.1})$$

we compute both marginal and multivariate diagnostics.

Finite-sample caveats. Marginal skewness/kurtosis estimates can be noisy when the subset size n is small, and multivariate diagnostics based on Mahalanobis radii require $n > d$ for a well-conditioned full-rank covariance estimate. Accordingly, we interpret large deviations most strongly on dataset–class subsets with ample sample sizes, and we treat results for very small subsets as qualitative indicators rather than definitive evidence of non-Gaussianity.

For each coordinate $j \in \{1, \dots, d\}$, let $\hat{\gamma}_{1,j}$ and $\hat{\gamma}_{2,j}$ denote the bias-corrected sample skewness and Pearson kurtosis, respectively, so that the Gaussian reference is $\hat{\gamma}_{1,j} = 0$ and $\hat{\gamma}_{2,j} = 3$. We summarize these marginal statistics by their median and interquartile range across dimensions, and we report the proportion of dimensions satisfying the loose Gaussian screen

$$\text{PctClose} = \frac{100}{d} \sum_{j=1}^d \mathbf{1}(|\hat{\gamma}_{1,j}| < 0.5 \wedge 2 < \hat{\gamma}_{2,j} < 5). \quad (\text{G.2})$$

At the multivariate level, we compute the empirical squared Mahalanobis radii

$$\delta_i^2 = (z_i - \bar{z})^\top \hat{\Sigma}^{-1} (z_i - \bar{z}), \quad (\text{G.3})$$

and the normalized Mardia kurtosis

$$\tilde{\beta}_{2,d} = \frac{1}{d(d+2)} \cdot \frac{1}{n} \sum_{i=1}^n (\delta_i^2)^2, \quad (\text{G.4})$$

whose Gaussian reference value is 1. For completeness, we also report

$$R_{\chi^2} = \frac{1}{nd} \sum_{i=1}^n \delta_i^2. \quad (\text{G.5})$$

R_{χ^2} should be interpreted with caution because $\hat{\Sigma}$ is estimated from the same sample, and therefore

$$\frac{1}{nd} \sum_{i=1}^n \delta_i^2 \approx \frac{n-1}{n} \quad (\text{G.6})$$

in the ideal full-rank unregularized setting, so values near 1 are largely expected and do *not* constitute strong evidence of Gaussianity.

To summarize the *second-order* geometry of each class-conditional feature cloud, let $\lambda_1 \geq \dots \geq \lambda_d$ denote the eigenvalues of $\hat{\Sigma}$ and define $p_j = \lambda_j / \sum_{k=1}^d \lambda_k$. We report the effective rank

$$E_{\text{rank}}(\hat{\Sigma}) = \exp\left(-\sum_{j=1}^d p_j \log p_j\right), \quad (\text{G.7})$$

which can be interpreted as the effective number of principal directions carrying variance, and the anisotropy

$$\text{Aniso}(\hat{\Sigma}) = \frac{\lambda_1}{\frac{1}{d} \sum_{j=1}^d \lambda_j} = \frac{d \lambda_1}{\text{tr}(\hat{\Sigma})}, \quad (\text{G.8})$$

which equals 1 for isotropic covariance and increases as variance concentrates into a small number of directions.

Table 23 | **Marginal Gaussianity diagnostics for class-conditional evaluation features.** For each dataset–class subset, we compute the bias-corrected sample skewness $\hat{\gamma}_{1,j}$ and Pearson kurtosis $\hat{\gamma}_{2,j}$ for every feature dimension j . We report the median with first/third quartiles [Q1,Q3] across dimensions, together with the percentage of dimensions satisfying the heuristic Gaussian screen $|\hat{\gamma}_{1,j}| < 0.5$ and $2 < \hat{\gamma}_{2,j} < 5$. These quantities are descriptive diagnostics rather than formal normality tests.

Dataset	Class	$\hat{\gamma}_{1,j}$	$\hat{\gamma}_{2,j}$	PctClose
Out-of-Distribution Existing Datasets with only Real Images				
Unsplash (uns)	Real	0.0045 [-0.07,0.08]	3.0577 [2.99,3.13]	99.77
InstagramImagesWithCaptions (ins)	Real	-0.0001 [-0.07,0.06]	3.0327 [2.97,3.01]	98.98
AnimeFace (Churchill and Chao, 2019)	Real	-0.0058 [-0.16,0.16]	3.1563 [3.01,3.30]	94.45
AnimeImages (Banana_Leopard, 2023)	Real	-0.0027 [-0.08,0.08]	3.0867 [3.02,3.17]	99.53
CelebA-Spoof (Zhang et al., 2020)	Real	-0.0081 [-0.12,0.12]	3.1013 [2.99,3.21]	99.38
Out-of-Distribution Existing Datasets with only Synthetic Images				
AGIQA-1k (Zhang et al., 2023)	Fake	-0.0034 [-0.13,0.11]	2.9904 [2.84,3.16]	97.81
AGIQA-3k (Li et al., 2024)	Fake	-0.0037 [-0.09,0.09]	3.0732 [2.97,3.19]	99.45
SPAI (Karageorgiou et al., 2025)	Fake	0.0018 [-0.06,0.06]	3.0477 [2.97,3.14]	99.69

Continued on next page

Dataset	Class	$\hat{y}_{1,j}$	$\hat{y}_{2,j}$	PctClose
SynthBuster Extended (Bamney, 2024; Guillaro et al., 2025)	Fake	-0.0004 [-0.08,0.07]	3.0470 [2.96,3.14]	99.38
SynthScars (Kang et al., 2025)	Fake	-0.0037 [-0.06,0.05]	3.0438 [2.98,3.11]	99.92
GigaGAN (Kang et al., 2023)	Fake	0.0016 [-0.08,0.08]	3.1237 [3.05,3.12]	100.00
LatentDiffusion (Corvi et al., 2023)	Fake	0.0043 [-0.08,0.09]	3.0735 [2.95,3.19]	99.45
MidjourneyV6 (ter)	Fake	0.0008 [-0.04,0.04]	3.0391 [2.99,3.09]	100.00
Co-Spy-Bench (Cheng et al., 2025)	Fake	-0.0005 [-0.05,0.05]	3.0572 [3.01,3.11]	100.00
Dalle3 (Egan et al., 2024)	Fake	0.0003 [-0.04,0.04]	3.0211 [2.98,3.06]	99.38
Out-of-Distribution Existing Datasets with both Real and Synthetic Images				
FourierSpectrumDiscrepancies (Dzanic et al., 2020)	Real	-0.0496 [-0.41,0.35]	2.4977 [2.01,3.16]	38.44
FourierSpectrumDiscrepancies (Dzanic et al., 2020)	Fake	0.0286 [-0.31,0.33]	2.9860 [2.60,3.52]	70.39
FakeInversion (Cazenavette et al., 2024)	Real	0.0027 [-0.08,0.09]	3.0182 [2.88,3.18]	99.69
FakeInversion (Cazenavette et al., 2024)	Fake	-0.0019 [-0.08,0.09]	3.0231 [2.89,3.18]	100.0
UniversalFakeDetect (Ojha et al., 2023)	Real	0.0019 [-0.06,0.06]	3.0278 [2.94,3.12]	99.22
UniversalFakeDetect (Ojha et al., 2023)	Fake	0.0021 [-0.07,0.07]	3.1456 [3.07,3.25]	99.84
DalleRecognition (Koliha, 2024)	Real	0.0025 [-0.07,0.08]	3.0176 [2.92,3.12]	99.61
DalleRecognition (Koliha, 2024)	Fake	-0.0002 [-0.04,0.05]	3.0393 [2.98,3.10]	99.77
Chameleon (Yan et al., 2024)	Real	0.0000 [-0.05,0.05]	3.0130 [2.96,3.07]	100.0
Chameleon (Yan et al., 2024)	Fake	-0.0073 [-0.09,0.08]	3.0824 [2.99,3.19]	99.84
AIGI-Detection-Quality-Paradox (Xiao et al., 2025)	Real	-0.0002 [-0.05,0.05]	3.0274 [2.96,3.12]	98.36
AIGI-Detection-Quality-Paradox (Xiao et al., 2025)	Fake	-0.0020 [-0.07,0.07]	3.0670 [2.99,3.15]	99.84
DiTFake (Li et al., 2025b)	Real	0.0054 [-0.05,0.06]	3.0002 [2.93,3.07]	99.84
DiTFake (Li et al., 2025b)	Fake	0.0043 [-0.07,0.08]	2.9940 [2.91,3.09]	100.0
Diffusion1kStep (Tan et al., 2024)	Real	0.0062 [-0.07,0.08]	3.0114 [2.92,3.11]	98.67
Diffusion1kStep (Tan et al., 2024)	Fake	-0.0010 [-0.16,0.17]	3.1901 [2.99,3.39]	97.27
GANGen-Detection (Chuangchuang Tan, 2024)	Real	-0.0022 [-0.10,0.11]	3.1389 [3.02,3.28]	97.27
GANGen-Detection (Chuangchuang Tan, 2024)	Fake	0.0091 [-0.17,0.19]	3.3372 [3.19,3.51]	94.38
RobustLDM (Rajan et al., 2024)	Real	0.0029 [-0.07,0.07]	3.0167 [2.93,3.11]	99.77
RobustLDM (Rajan et al., 2024)	Fake	-0.0015 [-0.10,0.10]	3.0445 [2.93,3.17]	99.38
RealRobustBench (Li et al., 2025a)	Real	0.0017 [-0.06,0.07]	3.0111 [2.94,3.08]	99.69
RealRobustBench (Li et al., 2025a)	Fake	0.0038 [-0.09,0.10]	3.0029 [2.90,3.13]	99.22
LDMFakeDetect (Rajan and Lee, 2025)	Real	0.0029 [-0.07,0.07]	3.0167 [2.93,3.11]	99.77
LDMFakeDetect (Rajan and Lee, 2025)	Fake	-0.0044 [-0.10,0.09]	3.0534 [2.95,3.19]	99.45
DIF (Sinita and Fried, 2024)	Real	0.0001 [-0.06,0.06]	3.0288 [2.97,3.09]	99.84
DIF (Sinita and Fried, 2024)	Fake	0.0032 [-0.09,0.10]	3.0732 [2.94,3.21]	99.53
ForenSynths (Wang et al., 2020)	Real	0.0143 [-0.23,0.24]	3.2898 [2.97,3.50]	97.27
ForenSynths (Wang et al., 2020)	Fake	0.0119 [-0.20,0.23]	3.1450 [2.84,3.40]	93.98
DNFTestSet (Zhang and Xu, 2025)	Real	0.0006 [-0.06,0.07]	3.0247 [2.95,3.10]	99.69
DNFTestSet (Zhang and Xu, 2025)	Fake	0.0038 [-0.11,0.13]	3.1119 [2.95,3.24]	99.45
DeepFakeFace (Song et al., 2023)	Real	-0.0021 [-0.06,0.06]	3.0008 [2.95,3.06]	99.84
DeepFakeFace (Song et al., 2023)	Fake	-0.0027 [-0.09,0.08]	3.0757 [3.01,3.15]	99.84
AIGCDetectBench (Zhong et al., 2024)	Real	0.0020 [-0.05,0.05]	3.0221 [2.97,3.07]	99.61
AIGCDetectBench (Zhong et al., 2024)	Fake	0.0079 [-0.09,0.10]	3.1043 [3.00,3.21]	99.61

Continued on next page

Dataset	Class	$\hat{\gamma}_{1,j}$	$\hat{\gamma}_{2,j}$	PctClose
AIGI-Holmes (Zhou et al., 2025b)	Real	0.0012 [-0.05,0.05]	3.0068 [2.95,3.06]	99.77
AIGI-Holmes (Zhou et al., 2025b)	Fake	0.0000 [-0.09,0.09]	3.0812 [2.99,3.17]	100.0
AI-Artwork (Kholy, 2024)	Real	-0.0032 [-0.06,0.06]	3.0122 [2.95,3.08]	99.77
AI-Artwork (Kholy, 2024)	Fake	0.0050 [-0.12,0.12]	3.0849 [2.95,3.23]	99.30
DiffusionForensics (Wang et al., 2023)	Real	0.0079 [-0.12,0.12]	3.1291 [3.02,3.22]	99.45
DiffusionForensics (Wang et al., 2023)	Fake	0.0071 [-0.31,0.32]	3.7335 [3.49,4.01]	71.48
B-Free (Guillaro et al., 2025)	Real	0.0049 [-0.05,0.06]	2.9950 [2.93,3.07]	99.92
B-Free (Guillaro et al., 2025)	Fake	0.0006 [-0.06,0.06]	3.0396 [2.98,3.10]	100.0
LASTED (Wu et al., 2025a)	Real	0.0078 [-0.06,0.07]	3.0478 [2.98,3.11]	100.0
LASTED (Wu et al., 2025a)	Fake	-0.0028 [-0.12,0.12]	3.2195 [3.11,3.34]	99.22
AIGIBench (Li et al., 2025c)	Real	0.0001 [-0.06,0.06]	3.0447 [2.99,3.11]	99.92
AIGIBench (Li et al., 2025c)	Fake	0.0067 [-0.11,0.13]	3.1626 [3.05,3.28]	99.77
DeepFakeBench (Yan et al., 2023)	Real	-0.0008 [-0.17,0.18]	3.1662 [2.97,3.34]	98.52
DeepFakeBench (Yan et al., 2023)	Fake	0.0101 [-0.16,0.17]	3.0508 [2.75,3.31]	94.69

Table 24 | **Multivariate diagnostics for class-conditional evaluation features.** Let $r_i = z_i - \bar{z}$ and let $\hat{\Sigma}$ be the regularized sample covariance of the class-conditional features. We compute squared Mahalanobis radii $\delta_i^2 = r_i^\top \hat{\Sigma}^{-1} r_i$. We report (i) the normalized Mardia kurtosis $\tilde{\beta}_{2,d} = \frac{1}{d(d+2)} \frac{1}{n} \sum_{i=1}^n (\delta_i^2)^2$, whose Gaussian reference value is 1, (ii) the normalized mean radius $R_{\chi^2} = \frac{1}{nd} \sum_{i=1}^n \delta_i^2$ (a limited self-consistency diagnostic; with same-sample mean/covariance estimation it is expected to be near 1 even under non-Gaussianity), and two spectrum/conditioning summaries of $\hat{\Sigma}$: (iii) the effective rank $E_{\text{rank}} = \exp(-\sum_{j=1}^d p_j \log p_j)$ with $p_j = \lambda_j / \sum_k \lambda_k$, and (iv) anisotropy $\text{Aniso} = \lambda_1 / (\frac{1}{d} \sum_{j=1}^d \lambda_j) = d\lambda_1 / \text{tr}(\hat{\Sigma})$, where $\lambda_1 \geq \dots \geq \lambda_d$ are the eigenvalues of $\hat{\Sigma}$. $E_{\text{rank}} \in [1, d]$ summarizes effective dimensionality, while $\text{Aniso} \geq 1$ measures dominance of the top variance direction. All quantities are descriptive diagnostics rather than calibrated hypothesis tests. (**Note:** We report dataset/classes where number of samples is greater than number of feature dimensions.)

Dataset	Class	$\tilde{\beta}_{2,d}$	R_{χ^2}	E_{rank}	Aniso
Out-of-Distribution Existing Datasets with only Real Images					
Unsplash (uns)	Real	1.1225	0.9998	227.80	75.23
InstagramImagesWithCaptions (ins)	Real	1.1092	0.9998	251.47	74.98
AnimeFace (Churchill and Chao, 2019)	Real	1.1957	0.9995	61.89	293.73
AnimeImages (Banana_Leopard, 2023)	Real	1.1715	0.9998	254.45	61.78
CelebA-Spoof (Zhang et al., 2020)	Real	1.1694	0.9998	171.74	166.86
Out-of-Distribution Existing Datasets with only Synthetic Images					
AGIQA-3k (Li et al., 2024)	Fake	1.0319	0.9992	210.56	76.71
SPAI (Karageorgiou et al., 2025)	Fake	1.0574	0.9994	267.96	51.12
SynthBuster Extended (Bammey, 2024; Guillaro et al., 2025)	Fake	1.0879	0.9996	194.50	91.25
SynthScars (Kang et al., 2025)	Fake	1.0690	0.9997	250.11	60.05
GigaGAN (Kang et al., 2023)	Fake	1.1644	0.9998	268.03	80.58
LatentDiffusion (Corvi et al., 2023)	Fake	1.1179	0.9997	212.48	76.55
MidjourneyV6 (ter)	Fake	1.0934	0.9998	281.47	50.06
Co-Spy-Bench (Cheng et al., 2025)	Fake	1.0772	0.9998	299.09	39.92
Dalle3 (Egan et al., 2024)	Fake	1.0995	0.9998	280.66	54.19

Continued on next page

Dataset	Class	$\tilde{\beta}_{2,d}$	R_{χ^2}	E_{rank}	Aniso
Out-of-Distribution Existing Datasets with both Real and Synthetic Images					
UniversalFakeDetect (Ojha et al., 2023)	Real	1.0030	0.9991	339.72	76.99
UniversalFakeDetect (Ojha et al., 2023)	Fake	1.0742	0.9996	282.82	52.51
DalleRecognition (Koliha, 2024)	Real	1.0335	0.9995	229.58	111.29
DalleRecognition (Koliha, 2024)	Fake	1.0719	0.9997	291.76	39.74
Chameleon (Yan et al., 2024)	Real	1.0619	0.9997	262.78	77.87
Chameleon (Yan et al., 2024)	Fake	1.1232	0.9996	194.40	119.96
AIGI-Detection-Quality-Paradox (Xiao et al., 2025)	Real	1.0430	0.9995	319.79	61.35
AIGI-Detection-Quality-Paradox (Xiao et al., 2025)	Fake	1.1087	0.9997	263.10	55.39
DiTFake (Li et al., 2025b)	Real	1.0507	0.9997	281.90	51.76
DiTFake (Li et al., 2025b)	Fake	1.0900	0.9997	213.86	58.78
Diffusion1kStep (Tan et al., 2024)	Real	1.0448	0.9997	222.71	109.72
Diffusion1kStep (Tan et al., 2024)	Fake	1.2005	0.9997	148.31	148.77
GANGen-Detection (Chuangchuang Tan, 2024)	Real	1.1642	0.9997	187.29	101.55
GANGen-Detection (Chuangchuang Tan, 2024)	Fake	1.2674	0.9996	146.67	122.43
RobustLDM (Rajan et al., 2024)	Real	1.0408	0.9996	249.19	98.38
RobustLDM (Rajan et al., 2024)	Fake	1.2930	0.9997	177.50	72.18
RealRobustBench (Li et al., 2025a)	Real	1.0891	0.9998	239.64	73.09
RealRobustBench (Li et al., 2025a)	Fake	1.1513	0.9997	173.27	98.95
LDMFakeDetect (Rajan and Lee, 2025)	Real	1.0408	0.9996	249.19	98.38
LDMFakeDetect (Rajan and Lee, 2025)	Fake	1.3017	0.9997	178.20	71.55
DIF (Sinitisa and Fried, 2024)	Real	1.0587	0.9998	309.16	60.52
DIF (Sinitisa and Fried, 2024)	Fake	1.1797	0.9997	192.48	89.98
ForenSynths (Wang et al., 2020)	Real	1.2396	0.9998	182.97	193.47
ForenSynths (Wang et al., 2020)	Fake	1.2685	0.9997	104.62	218.03
DNFTestSet (Zhang and Xu, 2025)	Real	1.0450	0.9997	274.26	84.16
DNFTestSet (Zhang and Xu, 2025)	Fake	1.1293	0.9997	198.64	117.17
DeepFakeFace (Song et al., 2023)	Real	1.0758	0.9998	237.47	74.91
DeepFakeFace (Song et al., 2023)	Fake	1.1205	0.9998	221.72	72.87
AIGCDetectBench (Zhong et al., 2024)	Real	1.0540	0.9998	323.16	69.08
AIGCDetectBench (Zhong et al., 2024)	Fake	1.1375	0.9997	228.13	83.67
AIGI-Holmes (Zhou et al., 2025b)	Real	1.0686	0.9998	303.84	49.64
AIGI-Holmes (Zhou et al., 2025b)	Fake	1.1375	0.9998	222.87	64.21
AI-Artwork (Kholy, 2024)	Real	1.0767	0.9998	220.19	71.57
AI-Artwork (Kholy, 2024)	Fake	1.2529	0.9997	144.42	114.62
DiffusionForensics (Wang et al., 2023)	Real	1.1251	0.9998	236.08	129.66
DiffusionForensics (Wang et al., 2023)	Fake	1.4326	0.9997	144.56	190.06
B-Free (Guillaro et al., 2025)	Real	1.0637	0.9998	278.64	52.92

Continued on next page

Dataset	Class	$\tilde{\beta}_{2,d}$	R_{χ^2}	E_{rank}	Aniso
B-Free (Guillaro et al., 2025)	Fake	1.0798	0.9998	267.28	55.60
LASTED (Wu et al., 2025a)	Real	1.0803	0.9998	300.73	58.42
LASTED (Wu et al., 2025a)	Fake	1.3136	0.9997	210.48	111.79
AIgIBench (Li et al., 2025c)	Real	1.0864	0.9998	303.42	65.94
AIgIBench (Li et al., 2025c)	Fake	1.1784	0.9998	217.27	77.97
DeepFakeBench (Yan et al., 2023)	Real	1.1376	0.9996	128.46	208.58
DeepFakeBench (Yan et al., 2023)	Fake	1.2176	0.9996	73.41	310.36

Applied to features from the strongest frozen encoder in our sweep, PE-Core-bigG-14-448 (Bolya et al., 2025), the diagnostics support Gaussian heads as approximate second-order models, but not as exact density models. In Table 23, most dataset–class subsets have median skewness near zero, median Pearson kurtosis near three, and high PctClose, often above 97%. The clearest marginal exceptions are the real and fake subsets of FourierSpectrumDiscrepancies (Dzanic et al., 2020) and the fake subset of DiffusionForensics (Wang et al., 2023), where deviations from the Gaussian reference are large.

For the Gaussian ladder, the multivariate diagnostics in Table 24 are more relevant. The normalized Mardia kurtosis $\tilde{\beta}_{2,d}$ is often only moderately above its Gaussian reference value of 1, but the larger deviations occur disproportionately in fake subsets. These same subsets frequently exhibit lower effective rank and higher anisotropy than their real counterparts, indicating that variance is concentrated in fewer dominant directions. This pattern is not universal, but it recurs often enough to make covariance-aware scoring plausible. By contrast, R_{χ^2} remains close to 1 almost everywhere, as expected under same-sample estimation, and should be interpreted only as a consistency check.

The resulting conclusion is deliberately narrow. Evaluation features are not exactly Gaussian, and several datasets exhibit clear departures from Gaussianity. For many dataset–class subsets, however, first- and second-order summaries describe a useful part of the class structure, while the failures mark cases where higher-order or multimodal structure likely matters. This supports Gaussian heads as controlled, interpretable probes in matched support-prior and encoder audits, not as literal generative models of the feature distribution.

Read together with the transfer results, these diagnostics support a methodological claim rather than a distributional one: when the support prior and frozen encoder feature space are fixed, Gaussian heads indicate the extent to which the observed head-level behavior is already available in low-order geometry. They do not imply that low-order geometry is the only transfer mechanism or that Gaussian assumptions hold globally. Head-level comparisons are therefore most interpretable when both quantities are controlled.